

Pythagorean–Burrau Nonperiodicity: Technical Details and Certificate Arguments

Working research draft

September 5, 2026

Abstract

This technical companion preserves the detailed reductions, asymptotic matching calculations, and certificate descriptions underlying the shorter main paper. The global conjecture remains open. The September 2026 review corrects the launch window in the validated-cover criterion and withdraws the unconditional terminal inference in the forced-lag argument. Numerical exploration and computer-assisted theorems have different evidential status; the individual certificate sources are indexed in the main paper and the repository review report.

Contents

1	Exact conjecture	2
2	Scaling and Euclid parameter	2
3	Second brakes and a quotient residual	2
4	Maximum-event reduction	3
5	Universal reduced system	7
6	Exact initial geometry	7
7	The real isosceles-right endpoint	9
8	The first skinny encounter	9
	8.1 The late-plunge restricted limit	10
	8.2 Reference phase and finite-parameter splitting	11
9	Cardinality and the remaining obstruction	37
10	Necessary integral identities	37
11	Open proof obligations	49

1 Exact conjecture

Let $a^2 + b^2 = c^2$ with positive integers, set $(m_1, m_2, m_3) = (a, b, c)$, and put

$$q_1 = (-c/2, 0), \quad q_2 = (c/2, 0), \quad q_3 = ((b^2 - a^2)/(2c), ab/c), \quad \dot{q}_i = 0.$$

The side opposite each mass has the same length as that mass. The conjecture asserts that no classical solution has a collision-free labelled period. Returns modulo rotation or label exchange and regularized continuation after collision are excluded.

2 Scaling and Euclid parameter

Lemma 1 (Simultaneous mass-length scaling). *If $q_i(t)$ solves the equations with masses m_i , then for $k > 0$*

$$\tilde{m}_i = km_i, \quad \tilde{q}_i(t) = kq_i(t/k)$$

solves the rescaled problem. A labelled period T corresponds exactly to a period kT .

Proof. Both sides of Newton's acceleration equation acquire the factor k^{-1} . Velocity is unchanged after the corresponding time substitution, and mutual distances acquire the positive factor k . $\square$

It follows that only primitive triples matter. For $u \in (0, 1)$ define

$$A(u) = \frac{1 - u^2}{1 + u^2}, \quad B(u) = \frac{2u}{1 + u^2}.$$

If $u = p/q$ is reduced, Euclid's integer triple is $(q^2 - p^2, 2pq, q^2 + p^2)$ and its common divisor is two exactly when p, q are both odd. Conversely Euclid's theorem recovers every ordered primitive triple. The inverse is $u = B/(1 + A)$, and leg exchange is the involution $u \mapsto (1 - u)/(1 + u)$. Its fixed point is $\sqrt{2} - 1$, giving the fundamental interval $0 < u \leq \sqrt{2} - 1$ for the real family.

After normalizing $c = 1$, masses and positions are

$$(A, B, 1), \quad q_1 = (-1/2, 0), \quad q_2 = (1/2, 0), \quad q_3 = ((B^2 - A^2)/2, AB).$$

3 Second brakes and a quotient residual

Lemma 2 (Second-brake lemma). *A classical free-fall segment generates a collision-free labelled periodic solution if and only if all inertial velocities vanish at some positive time before collision.*

Proof. Uniqueness and time reversal give $q(-t) = q(t)$. A brake at τ gives $q(\tau + s) = q(\tau - s)$ and hence a labelled period 2τ . Conversely, a labelled period T combined with the symmetry at zero gives reflection symmetry about $T/2$, and therefore zero velocity there. The period 2τ need not be minimal unless τ is the first positive brake. $\square$

Let $r_{ij} = |q_j - q_i|$, $K = \frac{1}{2} \sum_i m_i |\dot{q}_i|^2$, $U = \sum_{i < j} m_i m_j / r_{ij}$, and $H = K - U = -U_0$. In the center-of-mass frame define $\mu_1 = m_1 m_2 / (m_1 + m_2)$, $\mu_2 = m_3 (m_1 + m_2) / (m_1 + m_2 + m_3)$, and Jacobi vectors

$$X = q_2 - q_1, \quad Y = q_3 - \frac{m_1 q_1 + m_2 q_2}{m_1 + m_2}$$

and Hopf invariants

$$h(X, Y) = \left(\frac{|X|^2 - |Y|^2}{2}, X \cdot Y, X \times Y \right).$$

We use the quotient residual

$$\mathcal{B}(u, t) = \frac{d}{dt} h(X(u, t), Y(u, t)).$$

Away from triple collision, Dh has rank three and kernel equal to the common rotation direction. Since total linear and angular momentum vanish, $\mathcal{B} = 0$ is equivalent to vanishing of every labelled inertial velocity. Unlike three mutual-distance derivatives, this remains true at syzygy.

4 Maximum-event reduction

Let

$$I = \mu_1 |X|^2 + \mu_2 |Y|^2, \quad \sigma = \mu_1 \bar{X} \dot{X} + \mu_2 \bar{Y} \dot{Y}, \quad \zeta = \mu_1 \bar{X} \dot{X} - \mu_2 \bar{Y} \dot{Y}.$$

Here complex multiplication identifies the oriented plane with $\mathbb{C}$. Then $\dot{I} = 2 \operatorname{Re} \sigma$ and the angular momentum is $L = \operatorname{Im} \sigma$. Consequently, at $L = 0$ and away from $Y = 0$,

$$\dot{I} = 0 \text{ and } \zeta = 0 \iff \dot{X} = \dot{Y} = 0.$$

The implication from a brake to these three scalar equalities is unconditional. At the Jacobi degeneracy $Y = 0$, the Hopf residual $\mathcal{B}$ above supplies the coordinate-independent converse.

This splitting has an exact shape interpretation. Put $J_X = \mu_1 |X|^2$, $J_Y = \mu_2 |Y|^2$, $s = J_X/I$, and $\phi = \arg Y - \arg X$. At every zero of $\dot{I}$ with $X, Y \neq 0$ and $L = 0$,

$$\zeta = I \left(\dot{s} - 2is(1-s)\dot{\phi} \right), \quad |\zeta|^2 = \frac{8KJ_XJ_Y}{I} = 8KI s(1-s).$$

Indeed $\sigma = 0$ gives $\zeta = 2\mu_1 \bar{X} \dot{X} = -2\mu_2 \bar{Y} \dot{Y}$; decomposing each Jacobi velocity into radial and angular parts proves both formulas. Thus a maximum event is a brake exactly when its shape velocity vanishes, and the origin-avoidance problem can equivalently be written as the scalar strict inequality $K = U - U_0 > 0$. Because this scalar is nonnegative and has an even-order zero at a transverse brake, this reformulation does not create a sign-change argument by itself.

There is also a brake residual intrinsic to any binary Levi-Civita chart. For a selected pair write $g = q_j - q_i = w^2$, $dt = |w|^2 d\sigma$, and $z = dw/d\sigma$. Let G be the third body relative to the selected pair's center of mass and $P = \dot{G}$. With reduced masses μ_g, μ_G , angular momentum is

$$L = 2\mu_g \operatorname{Im}(\bar{w}z) + \mu_G G \times P.$$

Thus, on a collision-free zero-angular-momentum trajectory and wherever $G \neq 0$,

$$(z_r, z_i, G \cdot P) = 0 \iff \text{a labelled brake.}$$

Indeed $z = 0$ gives $\dot{g} = 0$; then $L = 0$ gives $G \times P = 0$, while $G \cdot P = 0$ forces $P = 0$. Total momentum removes the common translation. This residual stays regular through arbitrarily close selected-pair passages; at $G = 0$ one changes Jacobi tree or uses $\mathcal{B}$.

Theorem 3 (Brake-event identities). *Along a collision-free free-fall solution with initial potential U_0 , every brake satisfies*

$$K = 0, \quad U = U_0, \quad \ddot{I} = -2U_0 < 0, \quad r_{ij} \geq \frac{m_i m_j}{U_0}.$$

At any zero of $\dot{I}$ for which $\ddot{I} \geq 0$, one instead has $K \geq U_0$.

Proof. With positive potential magnitude U and energy $H = K - U = -U_0$, the Lagrange–Jacobi identity is

$$\ddot{I} = 4K - 2U = 2K - 2U_0 = 2U - 4U_0.$$

At a brake $K = 0$, giving the first three assertions. Since every summand $m_i m_j / r_{ij}$ is positive and bounded above by $U = U_0$, the separation bound follows. If $\ddot{I} \geq 0$, the middle expression instead gives $K \geq U_0$. $\square$

Corollary 4 (Validated-cover criterion). *Suppose a classical free-fall solution has a launch window $[0, t_L]$ on which $U < 2U_0$. Suppose validated flow enclosures cover every time in $[t_L, t_*]$, and every enclosure proves at least one of*

$$\dot{I} \neq 0, \quad K > 0, \quad \mathcal{B} \neq 0.$$

Then there is no second brake in $(0, t_]$. If the terminal enclosure satisfies a proved forward collision-or-escape criterion that excludes every later brake, the initial free-fall solution is nonperiodic.*

Proof. At launch $\dot{I}(0) = 0$, whereas $U < 2U_0$ implies $\ddot{I} < 0$. Consequently $\dot{I}(t) < 0$ for $0 < t \leq t_L$. The remaining disjunction excludes a brake at every covered time. This argument applies fiber by fiber when t_* depends on a parameter; both section images and the complete intervening flow tubes must be covered. The initial brake itself is never subjected to the nonvanishing disjunction. $\square$

This criterion has been implemented with exact rational initial data and MPFR interval arithmetic. The archived certificates prove the conjecture for

$$(21, 20, 29), (5311, 5280, 7489), (8319, 8200, 11681), (33439, 32400, 46561), (72, 65, 97)$$

in both leg orderings. These are five point theorems, not a parameter-cover argument.

Lemma 5 (Terminal escape certificate). *For a prospective binary of total mass M and an outer body of mass m_c , let r, ρ be the inner and outer Jacobi radii and $e = |\dot{x}|^2/2 - M/r$. Choose $\eta > 0$, set $R = M/\eta$ and $s_0 = \rho_0 - R$, and suppose*

$$s_0 > 0, \quad \dot{\rho}_0 > 0, \quad \delta = \frac{1}{2}\dot{\rho}_0^2 - \frac{M + m_c}{s_0} > 0,$$

$$e_0 + \frac{m_c \sqrt{2MR}}{\sqrt{2\delta} s_0^2} < -\eta.$$

Then the future solution either ends in inner binary collision or the outer body escapes with $\dot{\rho} \geq \sqrt{2\delta}$ forever. In particular there is no later classical brake.

Proof. On the bootstrap region $e < -\eta$, one has $r < R$ and $r|\dot{x}| < \sqrt{2MR}$. The Newton field derivative bound $\|D(v/|v|^3)\| \leq 2/|v|^3$ gives

$$|\dot{e}| \leq \frac{2m_c r |\dot{x}|}{(\rho - r)^3}.$$

Meanwhile

$$J = \frac{1}{2}\dot{\rho}^2 - \frac{M + m_c}{\rho - R}$$

is nondecreasing while $\dot{\rho} > 0$. Hence $\rho \geq \rho_0 + \sqrt{2\delta}(t - t_0)$, and the total future tidal increase of e is at most the allowance in the hypothesis. Its strict margin closes the bootstrap. $\square$

Theorem 6 (Validated middle first-maximum interval). *For every real*

$$\frac{29}{100} \leq u \leq \frac{14501}{50000},$$

the tied classical trajectory is collision-free through its first positive local maximum of I , and this maximum is not a labelled brake. The unique intervening minimum and the first subsequent maximum obey

$$t_{\min} \in [0.758038, 0.758045], \quad t_{\max} \in [1.32908, 1.33943].$$

Proof. The exact tied launch is embedded as a one-dimensional mean-value graph with u retained as a frozen state coordinate. A pinned MPFR/CAPD C^1 Poincare calculation propagates this graph first in a pair-13 Levi-Civita chart and, after an exact Jacobi-tree transformation at $t = 1$, in a pair-23 chart. On every accepted-step enclosure, all three mutual separations are strictly positive. The launch-concavity window and whole-leg audits give $\dot{I} < 0$ after launch to the first minus-to-plus zero and $\dot{I} > 0$ from that zero to the first later plus-to-minus zero. At the minimum, $U/U_0 \in [5.17942, 5.18335]$; at the maximum, $U/U_0 \in [0.983016, 1.01907] \subset (0, 2)$, proving both events strict by Lagrange–Jacobi. In the final selected-pair chart the validated residual has

$$\text{Im } z \in [-0.0500738, -0.0113187], \quad P_x \in [-0.0561842, -0.00917817].$$

A labelled brake forces $z = P = 0$, so either strict component excludes it. The exact chart identities are symbolically regressed. Two abutting tiles cover the stated interval and an independent replay reproduces the left-tile terminal PASS enclosure. The source, complete replay, and pinned dependency hash are archived with the paper. $\square$

This theorem excludes only the first positive maximum. It does not rule out a brake on a later maximum branch and therefore is not, by itself, a nonperiodicity theorem on the displayed interval.

Theorem 7 (Validated middle real interval). *For every real*

$$\frac{29}{100} \leq u \leq \frac{2900000101}{10000000000},$$

the tied maximal classical solution has no second labelled brake at any collision-free time and hence is not a classical periodic solution. Consequently the Pythagorean–Burrau conjecture holds for infinitely many primitive triples in this interval, including $(9159, 5800, 10841)$ at $u = 29/100$.

More explicitly, for every integer $k \geq 99010$ set

$$u_k = \frac{290k + 1}{1000k} = \frac{29}{100} + \frac{1}{1000k}.$$

The numerator and denominator are coprime and of opposite parity, and the interval inequality is equivalent to $1/(1000k) \leq 101/10^{10}$. The theorem therefore proves nonperiodicity for the primitive triples

$$(915900k^2 - 580k - 1, 580000k^2 + 2000k, 1084100k^2 + 580k + 1), \quad k \geq 99010.$$

Proof. A pinned MPFR/CAPD propagation embeds the exact launch curve on $[0.29, 0.2900000001]$ in one correlated tripleton with u frozen as a state coordinate. Before the exchange, every accepted full-step enclosure is checked for three positive mutual separations. The launch window verifies

$U < 2U_0$; subsequent windows verify the brake-exclusion disjunction of Corollary 4. Three algebraic changes of Levi–Civita chart use rigorous mean-value images.

Across the exchange, six C^1 Poincare maps project onto the transverse pair–13 sections

$$w_r = -\frac{3}{5}, -\frac{1}{2}, -\frac{2}{5}, -\frac{3}{10}, -\frac{1}{5}, 0.$$

For each map a convex mean-value domain contains the propagated family and its stored center, the validated section derivative transports the distinguished parameter generator, and an independent C^0 integration covers the complete incoming tube through the latest possible return. Thus section reconditioning does not omit any inter-section time. Every tube enclosure retains positive separation and excludes a brake.

For the terminal binary $\{2, 3\}$ set $M = B + 1$, $G = q_1 - C_{23}$, $\rho = |G|$, $h = |\dot{q}_3 - \dot{q}_2|^2/2 - M/r_{23}$, and

$$R = M/4, \quad d = \rho - R, \quad E_\rho = \frac{\dot{\rho}^2}{2} - \frac{A + B + 1}{d}, \quad \Delta = \frac{A\sqrt{2MR}}{\sqrt{2E_\rho}d^2}.$$

These are precisely the quantities of Lemma 5; h is transported as a regularized state variable. Hence the terminal check can use its enclosure without dividing by the small inner separation. The resulting terminal enclosure has

$$\begin{aligned} t_* &\in [4.30030837364, 4.30032429905], \\ d &> 1.8655, \quad \dot{\rho} > 2.0484, \quad E_\rho > 0.8223, \\ h &< -9.2745, \quad -4 - h - \Delta > 5.0690. \end{aligned}$$

These are the strict hypotheses of the phase-robust binary–escaper theorem with binary $\{2, 3\}$, escaper body 1, and $\eta = 4$. Hence after t_* the inner pair either collides classically, ending the maximal classical solution, or body 1 escapes permanently; neither alternative permits a later brake. The second-brake lemma proves nonperiodicity. A separately compiled interval checker verifies the terminal inequalities on an outward-widened copy of the printed box. The complete source, pinned dependency hash, per-section tube audits, and replay log accompany the proof. A second end-to-end run at 96-bit MPFR precision, tolerance 10^{-16} , and Taylor order 26 reproduces all six section audits, 3344 main-flow steps, and the terminal margin.

The remaining extension is the union of 100 independently certified, adjacent closed tiles of exact width 10^{-10} , beginning at 0.2900000001 and ending at 0.2900000101. Every requested interval is identical to its PASS interval, every consecutive pair shares its exact rational endpoint, and all 100 full logs are archived. The hardened base tile and the synchronized one-set theorem overlap this campaign, so their union is the complete closed interval stated above, with no topological or rounding gap. $\square$

This continuum theorem covers a width of 1.01×10^{-8} . Larger parameter covers remain a separate validation problem; experimental wider propagations are not included in its conclusion.

For each fixed u the separation bound is positive, and it is uniform on every compact parameter interval bounded away from $u = 0$. It degenerates in the skinny limit and is not a parameter-uniform Euclidean separation theorem on the whole open interval.

On any regular strict-maximum branch, the implicit-function theorem locally gives $t = t_k(u)$. The unresolved condition becomes the planar origin-avoidance problem

$$u \mapsto \zeta(u, t_k(u)) \in \mathbb{C} \setminus \{0\}.$$

To turn this into a universal theorem one must also control branch creation, collision and escape endpoints, degenerate events, and the possible number of branches. The nominal dimension count alone supplies none of these facts.

Ordinary signed-event diagnostics rule out a simpler global shortcut. On the single ($u=1/3$) trajectory, maximum-event values of (ζ_a) occupy all four quadrants, and three early values have the origin in their convex hull. Thus no fixed real linear projection of ζ can be sign-definite over all maximum events. A successful obstruction must instead use branch-wise order or winding, a nonlinear barrier, or a state-dependent projection.

5 Universal reduced system

Let

$$\mu_1 = \frac{m_1 m_2}{m_1 + m_2}, \quad \mu_2 = \frac{m_3(m_1 + m_2)}{m_1 + m_2 + m_3}.$$

Rotate so that $X = (R, 0)$ and write $Y = (P, Q)$. At zero angular momentum, Routh reduction gives

$$K_{red} = \frac{\mu_1 \dot{R}^2 + \mu_2 (\dot{P}^2 + \dot{Q}^2)}{2} - \frac{\mu_2^2 (P\dot{Q} - Q\dot{P})^2}{2[\mu_1 R^2 + \mu_2 (P^2 + Q^2)]}.$$

The positive potential is

$$U = \frac{m_1 m_2}{R} + \frac{m_1 m_3}{\sqrt{(P + \frac{m_2 R}{m_1 + m_2})^2 + Q^2}} + \frac{m_2 m_3}{\sqrt{(P - \frac{m_1 R}{m_1 + m_2})^2 + Q^2}}.$$

The Pythagorean curve on the brake boundary is

$$R = 1, \quad P = \frac{AB(B - A)}{A + B}, \quad Q = AB, \quad \dot{R} = \dot{P} = \dot{Q} = 0.$$

6 Exact initial geometry

The initial center of mass is the incenter because its barycentric weights are the opposite side lengths. With $I = \sum m_i |q_i - q_{CM}|^2$ and positive potential magnitude U , direct calculation gives

$$I_0 = abc, \quad U_0 = \frac{ab}{c} + \frac{ac}{b} + \frac{bc}{a}.$$

Thus after normalization

$$I_0 = AB, \quad U_0 = AB + (AB)^{-1}, \quad H = -U_0.$$

The configurational measure at fixed masses is

$$\mu_0 = U_0 \sqrt{I_0} = (AB)^{3/2} + (AB)^{-1/2},$$

and decreases strictly on the fundamental interval. It is not invariant under the simultaneous mass-length scaling used above.

For $D = r_{12}^2 - r_{23}^2 - r_{31}^2$, Newton's equations give

$$D(0) = \dot{D}(0) = 0, \quad \ddot{D}(0) = 2 \left(\frac{1}{AB} - AB(A+B) \right) > 0.$$

This is local only. Converged ordinary numerical experiments show D later changes sign for $u = 1/3$, so a global $D > 0$ cone is not pursued. An exact fourth-order calculation gives

$$D^{(4)}(0) = \frac{P_{20}(u)}{u^4(u-1)^4(u+1)^2(1+u^2)^4}.$$

For $t = 2u$, the 21 coefficients of $P_{20}(t/2)$ in the degree-20 Bernstein basis are

$$\begin{aligned} & -1, -\frac{19}{20}, -\frac{86}{95}, -\frac{1313}{1520}, -\frac{63901}{77520}, -\frac{3823}{4864}, -\frac{185677}{248064}, -\frac{1177527}{1653760}, \\ & -\frac{1819581}{2687360}, -\frac{27715771}{42997760}, -\frac{58252257}{94595072}, -\frac{101898611}{171991040}, -\frac{24804157}{42997760}, \\ & -\frac{181712839}{317521920}, -\frac{37060991}{63504384}, -\frac{157122943}{254017536}, -\frac{219023483}{317521920}, \\ & -\frac{10188109}{12451840}, -\frac{25856203}{24903680}, -\frac{3686723}{2621440}, -\frac{2112047}{1048576}. \end{aligned}$$

They are all strictly negative. Positivity and partition of unity of that basis prove $P_{20} < 0$ on $0 \leq u \leq 1/2$. Hence $D^{(4)}(0) < 0$ throughout the fundamental interval. This analytic uniform local sign still supplies no global cone.

One further exact recurrence gives

$$D^{(6)}(0) = -\frac{R_{30}(u)}{8u^7(1-u)^7(1+u)^3(1+u^2)^5}.$$

For $t = 2u$, the degree-30 Bernstein coefficients of $R_{30}(t/2)$ are

$$\begin{aligned} & 19, \frac{266}{15}, \frac{28861}{1740}, \frac{63103}{4060}, \frac{6395887}{438480}, \frac{7813165}{570024}, \frac{97995461}{7600320}, \\ & \frac{395397937}{32572800}, \frac{1903835447}{166483200}, \frac{822381181}{76304800}, \frac{44652539959}{4395156480}, \frac{1062904231}{110988800}, \\ & \frac{46339562047}{5134438400}, \frac{57924715549}{6813004800}, \frac{73076017417}{9128755200}, \frac{34682896469}{4604067840}, \frac{147143231691}{20763443200}, \\ & \frac{13083275107261}{142418163589111}, \frac{4249318181351}{177734203151437}, \\ & \frac{1962145382400}{1849150776667}, \frac{22673679974400}{43499899224473}, \frac{716010946560}{3826743850991}, \frac{31504481648640}{18951324099991}, \\ & \frac{340957593600}{644065041443}, \frac{8182982246400}{14565220793867}, \frac{711563673600}{178980837823}, \frac{3320630476800}{1746818009353}, \\ & \frac{99618914304}{99618914304}, \frac{1839118417920}{1839118417920}, \frac{17028874240}{17028874240}, \frac{116769423360}{116769423360}, \\ & \frac{2824577023}{125829120}, \frac{37475266287}{1073741824}. \end{aligned}$$

They are all positive, so $D^{(6)}(0) < 0$ throughout the fundamental interval. The finite pattern $D^{(2)} > 0, D^{(4)} < 0, D^{(6)} < 0$ supplies no global remainder estimate.

The same calculation for twice oriented area $\Delta_2 = (q_2 - q_1) \times (q_3 - q_1)$ gives

$$\ddot{\Delta}_2(0) = \frac{Q_{10}(u)}{2u^2(u-1)^2(1+u^2)^3},$$

where

$$Q_{10} = u^{10} - 10u^9 + 19u^8 - 28u^7 - 14u^6 + 8u^5 - 2u^4 - 4u^3 - 3u^2 + 2u - 1.$$

Put $t = 2u$. The eleven coefficients of $Q_{10}(t/2)$ in the degree-10 Bernstein basis are

$$\left(-1, -\frac{9}{10}, -\frac{49}{60}, -\frac{181}{240}, -\frac{241}{336}, -\frac{179}{252}, -\frac{991}{1344}, -\frac{3091}{3840}, -\frac{10693}{11520}, -\frac{2919}{2560}, -\frac{1543}{1024}\right).$$

They are all strictly negative. Positivity and partition of unity of the Bernstein basis prove $Q_{10} < 0$ on $0 \leq u \leq 1/2$, and hence $\ddot{\Delta}_2(0) < 0$ on the full fundamental interval. This proves uniform initial motion toward syzygy, not global area concavity.

7 The real isosceles-right endpoint

Theorem 8. *At $u = \sqrt{2} - 1$, the classical solution has no positive second brake and reaches a binary or triple collision in finite time.*

Proof. Here $m_1 = m_2 = m = 1/\sqrt{2}$ and reflection symmetry persists. Write the base bodies as $(-x, y_b), (x, y_b)$ and the apex as $(0, y_a)$, with $h = y_a - y_b$. Then

$$\ddot{x} = -\frac{m}{4x^2} - \frac{x}{(x^2 + h^2)^{3/2}} < 0.$$

Starting with $x_0 = 1/2$ and $\dot{x}(0) = 0$ gives $\dot{x} < 0$, excluding a second brake. While $x > 0$, one has $\ddot{x} \leq -m/(4x_0^2)$, whence

$$x(t) \leq x_0 - \frac{m}{8x_0^2}t^2.$$

Thus collision occurs no later than $\sqrt{8x_0^3/m} = 2^{1/4}$. $\square$

The endpoint is irrational, so this theorem does not settle an integer case. Its perturbation is singular because the collision unfolds into a close ordinary encounter when reflection symmetry is broken.

8 The first skinny encounter

Put $\epsilon = B$, $A = \sqrt{1 - \epsilon^2}$, and $M = A + 1$. For the tight-pair and outer Jacobi vectors

$$X = q_3 - q_1, \quad Y = q_2 - \frac{Aq_1 + q_3}{M},$$

rotate the encounter frame so that $X(0)/\epsilon = 1$ and scale

$$X = \epsilon x, \quad t = \frac{\epsilon^{3/2}}{\sqrt{M}}\tau.$$

The exact inner equation is

$$x_{\tau\tau} = -\frac{x}{|x|^3} + \frac{\epsilon^3}{M} \left[\Phi\left(y - \frac{A\epsilon}{M}x\right) - \Phi\left(y + \frac{\epsilon}{M}x\right) \right], \quad \Phi(v) = \frac{v}{|v|^3}.$$

The outer acceleration in τ -time is $O(\epsilon^3)$ on a fixed encounter interval. Introduce Levi-Civita variables

$$x = z^2, \quad d\tau = |z|^2 ds.$$

The augmented equations for z , z_s , the osculating Kepler energy, and the outer state are real analytic through $z = 0$. At $\epsilon = 0$,

$$z_0(s) = \cos(s/\sqrt{2}), \quad s_c = \pi/\sqrt{2}.$$

The first transverse parameter variation occurs at order five and satisfies

$$v_{ss} + \frac{1}{2}v = -\frac{3}{8}\cos^5(s/\sqrt{2}), \quad v(0) = v_s(0) = 0,$$

so that

$$v(s_c) = -\frac{15\pi}{128}.$$

Theorem 9 (First skinny miss). *There is $\epsilon_0 > 0$ such that for every real member with $0 < B = \epsilon < \epsilon_0$, the first tight-pair encounter is collision-free. At closest passage,*

$$r_{13,\min} = \frac{225\pi^2}{16384}\epsilon^{11}(1 + O(\epsilon)), \quad h_{13} = -\frac{15\pi}{64}\epsilon^{11/2} + O(\epsilon^{13/2}),$$

where $h_{13} = X \times \dot{X}$ is specific pair angular momentum.

Proof. Analytic dependence gives

$$\text{Im } z(s_c, \epsilon) = -\frac{15\pi}{128}\epsilon^5 + O(\epsilon^6).$$

Since $\partial_s \text{Re } z_0(s_c) = -1/\sqrt{2}$, choose a fixed neighborhood on which the real part remains strictly decreasing. Its unique zero has the displayed nonzero imaginary part. Outside that neighborhood z_0 is bounded away from zero up to a fixed outgoing section, while both outer separations are uniformly bounded below. Thus no collision occurs. Squaring the transverse displacement and converting the scaled angular momentum to physical time gives the two coefficients. $\square$

The first-encounter result alone does not prove skinny nonperiodicity. The binary remains bound with specific energy $-M/\epsilon + O(\epsilon^3)$, and the first-passage outer radial escape margin is $-2 + O(\epsilon)$. The outer body falls back on an order-one time scale, during which the binary executes $O(\epsilon^{-3/2})$ cycles. Later sections control this train and prove nonperiodicity on infinitely many phase windows, but not uniformly in phase.

8.1 The late-plunge restricted limit

The terminal lemma cannot be applied uniformly over the later binary phase. Indeed, rescale near the outer plunge by

$$X = \epsilon R, \quad Y = \epsilon Z, \quad t = t_* + \epsilon^{3/2}\theta.$$

On compact nonsingular sets the exact Jacobi equations tend to

$$R_{\theta\theta} = -2\frac{R}{|R|^3}, \quad Z_{\theta\theta} = -[\Phi(Z + R/2) + \Phi(Z - R/2)]. \quad (\text{LR})$$

After regularizing the prescribed binary, this convergence extends through its binary collisions provided the light particle is separate. The invariant rectilinear subspace has

$$R = (r, 0), \quad Z = (0, z), \quad \ddot{z} = -\frac{2z}{(z^2 + r^2/4)^{3/2}},$$

where r is the periodic collision–ejection binary with apocenter one. The outer instantaneous energy obeys

$$\frac{d}{d\theta} \left(\frac{1}{2} \dot{z}^2 - \frac{2}{\sqrt{z^2 + r^2/4}} \right) = \frac{r\dot{r}}{2(z^2 + r^2/4)^{3/2}}.$$

Theorem 10 (Restricted parabolic separatrix). *When the light particle crosses $z = 0$ at binary apocenter, there is a launch speed $v_* > 0$ for which it has no light–heavy collision and*

$$z(\theta) \rightarrow +\infty, \quad \dot{z}(\theta) \rightarrow 0$$

as $\theta \rightarrow +\infty$. Reflection and time reversal give a parabolic-to-parabolic scattering orbit.

Proof. For positive z , $\ddot{z} < 0$. Small launch speeds turn: with $r = \cos^2 \psi$ and $d\theta = \cos^2 \psi d\psi$, the zero-speed variational equation becomes, after $\zeta = \cos \psi w$,

$$w_{\psi\psi} + (1 + 14 \sec^2 \psi)w = 0.$$

Sturm comparison forces a simple zero before the first binary collision. Large launch speeds reach $z = 1$ while retaining a speed for which $\dot{z}^2/2 - 2/z > 0$; this quantity is thereafter nondecreasing. Let $\mathcal{T}$ be the nonempty open bounded set of speeds with a finite turn and put $v_* = \sup \mathcal{T}$. The boundary solution cannot turn and cannot have positive limiting speed, since either property is open. It cannot remain bounded, because the periodic attraction would then force a turn. $\square$

The binary collisions in (LR) are regularized features of an auxiliary limit, not physical continuations of a positive- ϵ orbit. Nevertheless the separatrix proves that a strict phase-uniform escape margin is impossible in the limiting problem. A finite- ϵ proof must instead match the incoming data and determine the separatrix splitting.

8.2 Reference phase and finite-parameter splitting

Put $A = \sqrt{1 - \epsilon^2}$, $M = 1 + A$, $N = M + \epsilon$, and let $\rho_0^2 = 2A^2/M$ be the exact initial outer Jacobi radius. Replacing the tight pair by a point mass gives the exact reference clock

$$\Phi_{\text{ref}} = \frac{\pi}{\epsilon^{3/2}} \sqrt{\frac{M}{N}} \rho_0^{3/2} = \pi \epsilon^{-3/2} - \frac{\pi}{4} \epsilon^{-1/2} - \frac{15\pi}{32} \epsilon^{1/2} + O(\epsilon^{3/2}). \quad (\text{RC})$$

Moreover $\Phi'_{\text{ref}} \sim -(3\pi/2)\epsilon^{-5/2}$. Thus every open phase arc has infinitely many reference preimage intervals accumulating at zero, each containing rational Euclid parameters. The matching theorem below transfers this reference statement to compact open windows of the true flow.

That long-cycle step can be proved without finite-eccentricity averaging. Fix $0 < \alpha < 1/6$ and stop at physical outer radius $\rho = \epsilon^\alpha$. Taylor cancellation in the outer Jacobi field gives a monopole remainder $O(\epsilon^2 \rho^{-4})$. Inner-energy and radial comparison estimates keep $|X| \leq 2\epsilon$ and give

$$t_\alpha - t_0(\epsilon^\alpha) = O(\epsilon^{2-3\alpha}).$$

In LC variables the scaled binary forcing is $O(\epsilon^{4-3\alpha})$ over regularized length $O(\epsilon^{-3/2})$. Harmonic-oscillator variation and the exact LC clock identity yield a regularized mean-anomaly error $O(\epsilon^{1-3\alpha})$. Hence, unless a prior classical collision has already ended the orbit, the renormalized incoming intercept satisfies

$$\text{dist}_{S^1}(\chi_\alpha, \Phi_{\text{ref}}) = O(\epsilon^{1/2-3\alpha}) = o(1). \quad (\text{MATCH})$$

Theorem 11 (Incoming-tail matching). *Fix a sufficiently large scaled incoming section Σ_K^- . As $\epsilon \rightarrow 0$, either a classical collision occurs before section reach or the exact regularized state $S_\epsilon(K)$ satisfies*

$$\text{dist}(S_\epsilon(K), \Gamma_K^-(\Phi_{\text{ref}}(\epsilon))) = o(1), \quad (\text{IT})$$

where $\Gamma_K^-(\chi)$ is the restricted incoming parabolic state with asymptotic mean-anomaly intercept χ .

Proof. Put $Y_\epsilon = \epsilon^{\alpha-1}$ with $0 < \alpha < 1/6$. The exact cancellation of the outer dipole leaves an $O(Y^{-4})$ remainder on the tail. If

$$\mathcal{H}_N = \frac{1}{2}|Z_\theta|^2 - \frac{N}{Y}, \quad \mathcal{L} = Z \times Z_\theta,$$

then the pre-plunge rates and a simultaneous energy-torque bootstrap give

$$|\mathcal{H}_N - E_{\text{ref},\epsilon}| \leq CY^{-3}, \quad |\mathcal{L}| \leq CY^{-3/2}, \quad -Y_\theta \asymp Y^{-1/2}. \quad (\text{TB})$$

These estimates are uniform for $K \leq Y \leq Y_\epsilon$.

Collision-uniform binary control uses exact LC first integrals rather than a singular Delaunay chart. With normalized energy E set

$$\mathcal{J} = (E + 1, L, K_{12}), \quad L = z_x p_y - z_y p_x, \quad K_{12} = p_x p_y - \frac{E}{2} z_x z_y.$$

On the radial LC circle,

$$\det \frac{\partial(L, K_{12})}{\partial(z_y, p_y)} = -\frac{1}{2}.$$

The three components are antipodal-invariant uniform normal coordinates, and direct differentiation of the forced LC equations gives $|\mathcal{J}_s| \leq C\epsilon Y^{-3}$. Complete LC blocks have uniformly comparable physical lengths; the two incomplete endpoint blocks are bounded directly in LC time. Consequently

$$|\mathcal{J}(Y)| \leq O(\epsilon^{5/2-3\alpha}) + C\epsilon Y^{-3/2},$$

and the accumulated binary-clock error is

$$O(\epsilon^{1-3\alpha/2}) + O(\epsilon |\log \epsilon|) = o(1). \quad (\text{BC})$$

Retain the exact finite- ϵ subtraction

$$F_\epsilon(Y) = n_{\text{bin}}(t_{\text{mono}} - t_0(\epsilon Y))$$

until Y is fixed. The reciprocal-speed consequence of (TB) and (BC) is

$$\text{dist}_{S^1}(\Lambda_\epsilon(Y) + F_\epsilon(Y), \chi_\epsilon) \leq CY^{-1/2} + o(1), \quad (\text{PG})$$

where $\chi_\epsilon = \Phi_{\text{ref}}(\epsilon) + o(1)$. The $CY^{-1/2}$ term retains the restricted quadrupole flight correction.

For any sequence $\epsilon_n \rightarrow 0$, LC compactness gives a diagonal limiting restricted tail. The inherited angular bounds are $Z_\perp = O(Y^{-1})$ and $Z'_\perp = O(Y^{-5/2})$. Reflection oddness factors the exact transverse equation as

$$\xi'' + (2Y^{-3} + O(Y^{-5})) \xi = 0.$$

With $x = \log Y$ this is an L^1 perturbation of a constant system whose modes are Y and $Y^{1/2}$; the inherited decay forces $\xi \equiv 0$. The limit is therefore the rectilinear incoming parabolic orbit. Taking $\epsilon_n \rightarrow 0$ at fixed Y in (PG), and only then $Y \rightarrow \infty$, identifies its intercept as the limit of $\Phi_{\text{ref}}(\epsilon_n)$. Uniqueness of the incoming McGehee graph proves (IT). $\square$

On the limiting symmetric orbit let $d^2 = z^2 + r^2/4$. Expansion of the exact finite-parameter Jacobi field gives

$$\begin{aligned} r'' &= -\frac{2}{r^2} - \epsilon \frac{r}{d^3} + \frac{\epsilon^2}{2r^2} + O(\epsilon^3), \\ Z_y'' &= -\frac{2z}{d^3} - \epsilon \frac{z}{d^3} + \epsilon^2 \frac{z}{2d^3} + O(\epsilon^3), \quad Z_x'' = \epsilon^2 \frac{3r^3}{16d^5} + O(\epsilon^3). \end{aligned} \quad (\text{FS})$$

The centered order- ϵ energy-transfer variation cancels by parity; longitudinal bending starts at order ϵ^2 . The first symmetry-allowed splitting direction is binary phase. Its naive whole-line Jacobi field is nevertheless gauge-degenerate: choosing the global time-translation field $\zeta = z'$ makes the proposed integrand exactly

$$\frac{d}{d\theta} \left(\frac{rr'}{2d^3} \right).$$

Because $z' = O(|\theta|^{-1/3})$ at parabolic infinity, zero limiting data do not eliminate this mode. A well-posed transversality theorem must instead fix the parabolic epoch or compare incoming and outgoing parabolic curves V_u, V_s on $z = 0$. Reversibility gives $V_s(\phi) = V_u(-\phi)$, so the target is $2V_u'(0) \neq 0$. Ordinary finite-cutoff calculations support this with a slope difference near -0.90623 . This numerical value is not certified; the following finite certificate proves the required nonzero sign.

There is a finite sufficient certificate for this transversality. Let $h'' = ah$, $h(0) = v_*$, $h'(0) = 0$, with $a = (4z^2 - r^2/2)/(z^2 + r^2/4)^{5/2}$. Fixed-softening comparison and an explicit early escape estimate give

$$\sqrt{8} < v_* < 4.$$

If $W = hz'' - h'z'$, then $W' = hbr'$ with $b = 3zr/[2(z^2 + r^2/4)^{5/2}]$. During the first binary contraction this proves $(h/z')' > 0$ as long as $h > 0$, and hence rules out a first zero of h . A force bound also gives

$$z(\pi/8) > \frac{\pi\sqrt{2}}{4} - \frac{2\pi^2}{27\sqrt{3}} > \frac{1}{2\sqrt{2}},$$

so $a > 0$ thereafter. Thus $h'(\pi/4) > 0$ forces h to grow forever.

This finite sign is now rigorous. Put $q = h/v$, $w = z'$, $p = q'$, and use eccentric anomaly $r = \cos^2 \psi$, $d\theta = r d\psi$. The regular compact system is

$$z_\psi = rw, \quad w_\psi = -\frac{2zr}{d^3}, \quad q_\psi = rp, \quad p_\psi = r \frac{4z^2 - r^2/2}{d^5} q, \quad (\text{CJ})$$

with $(z, w, q, p)(0) = (0, v, 1, 0)$ and terminal point $\psi = \pi/2$.

Theorem 12 (Computer-assisted finite Jacobi sign). *For every $v \in [14/5, 4]$, the solution of (CJ) exists through $\psi = \pi/2$ and*

$$p(\pi/2) > \frac{1}{125}.$$

In particular, the parabolic-separatrix field satisfies

$$h'(\pi/4) > \frac{14}{625} > 0.$$

Certificate description. CAPD 6.1.0 native outward-rounded intervals were used at the pinned commit 731079217a9254ea2948d742df2b170895effe7f. The interval $[14/5, 4]$ was divided into 256 overlapping slabs and each slab was propagated independently by an order-20 interval Taylor

solver with a doubleton set. Every terminal interval had lower endpoint greater than $1/125$. The complete verifier prints exact hexadecimal binary64 endpoints; the dependency check, concise certificate, and symbolic regression test are included in the repository. Since $14/5 < \sqrt{8} < v_* < 4$ and $h = v_*q$, the stated corollary follows. $\square$

Theorem 13 (Restricted parabolic transversality). *The incoming and outgoing parabolic curves of the rectilinear equal-heavy-mass restricted problem cross transversely at the reversible apocenter-centered orbit.*

Proof. The collision-regularized stable-manifold implication can be checked directly. Compactify infinity by $z = 2/x^2$. In eccentric anomaly the field becomes

$$x_\psi = -\frac{r}{4}x^3w, \quad w_\psi = -\frac{r}{2}x^4(1 + r^2x^4/16)^{-3/2}.$$

It is analytic at $(x, w) = (0, 0)$ and through $r = 0$. Since $\int_0^\pi r d\psi = \pi/2$, its time- π map has, after $X = x$, $Y = w - \sqrt{2}x$, the form

$$X_1 = X - \frac{\pi}{8}X^3(Y + \sqrt{2}X) + O_7, \quad Y_1 = Y + \frac{\pi\sqrt{2}}{8}X^3Y + O_7. \quad (\text{PM})$$

Before this shear, $y = w/\sqrt{2}$ puts the exact map in the factored form

$$x_1 = x - kx^3(y + R_1), \quad y_1 = y - kx^3(x + R_2), \quad k = \frac{\pi\sqrt{2}}{8},$$

where analyticity of the exact flow and the x^3, x^4 factors in the vector field make R_1, R_2 analytic of order at least four (write $x = x_0u$ to factor the exact flow by x_0^3 and x_0^4). The degree-four term obeys $p_1(X, 0) < 0$, $p_2(X, 0) = 0$, and $\partial_Y p_2(X, 0) > 0$ for $X > 0$. McGehee's theorem [20] therefore gives the fixed-phase parabolic stable curve $Y = \varphi(X)$ with $\varphi(0) = \varphi'(0) = 0$, analytic for $X > 0$. McGehee's graph construction supplies an arbitrarily small Lipschitz constant on a narrowed sector, hence $|\varphi'|$ is correspondingly small for $X > 0$.

Restricting (PM) to this graph gives, for positive constants c_i and small X ,

$$c_1X^4 \leq X - F(X) \leq c_2X^4, \quad 0 < F'(X) \leq 1 - c_3X^3.$$

Thus $X_n^{-3} \asymp n$, $\sum X_n^3 = \infty$, and every stable-curve tangent $\delta X_n = (F^n)'(X_0)\delta X_0$ tends to zero.

Take both section speeds positive in the direction from $z < 0$ to $z > 0$. The reversor $\tilde{z}(t) = -z(-t)$ then gives $V_s(\phi) = V_u(-\phi)$. Tangency at zero would force both slopes to vanish. In extended variables $(\delta z, \delta w, \delta \phi)$, the global time-shift and pinned stable-phase tangents are explicitly

$$T_{\text{time}} = (v_*, 0, 1), \quad T_{\text{pin}} = (0, V'_s(0), 1) = (0, 0, 1).$$

Their difference is the fixed-phase stable tangent with z -component $h = z' - \eta_{\text{pin}}$. The validated theorem and $a > 0$ imply $h(\theta) \geq h(\pi/4) + (14/625)(\theta - \pi/4)$. But

$$\delta X_n = -\frac{X_n^3}{4}h(\theta_n), \quad \theta_n = \theta_0 + n\pi/2,$$

so $X_n^{-3} \asymp n$ makes $\liminf |\delta X_n| > 0$, contradicting tangent contraction. $\square$

Theorem 14 (Restricted phase windows). *There are one-sided open binary-phase intervals adjacent to the centered separatrix such that the incoming parabolic orbit escapes hyperbolically on one interval and turns at finite positive z on the other.*

Proof. The analytic section difference $\Delta(\phi) = V_u(\phi) - V_s(\phi)$ is odd and has $\Delta'(0) \neq 0$. It therefore has a strict linear sign on the two sides of zero. McGehee's stable curve separates the local hyperbolic and returning sectors. On the returning side $z'' < 0$ for $z > 0$; any trajectory that neither turns nor escapes with positive speed would be another parabolic point on the unique stable curve. Thus it turns in finite time. $\square$

The finite-turn side contains a singular sequence that prevents a uniform restricted no-brake argument.

Theorem 15 (Restricted turn resonances). *For every sufficiently large integer n , there is a finite-turn phase $\phi_n \rightarrow 0$ whose first outer turn is simultaneous with the n th heavy-binary apocenter. It is a nonsingular brake state of the collision-regularized restricted flow. If $Z_{t,n}$ and T_n are its turn height and time, then*

$$Z_{t,n} = (2n)^{2/3}(1 + o(1)), \quad T_n = \frac{\pi n}{2} + o(1).$$

Proof. At center phase ϕ , the binary mean anomaly is $\Lambda(\theta) = \phi + 4\theta$. The first-turn time $T(\phi)$ is continuous on the finite-turn side and tends to infinity at the transverse parabolic boundary; otherwise finite-time convergence would give a turn on the parabolic separatrix. Thus $\phi + 4T(\phi)$ crosses $2\pi n$ for every large n . For $Z = Z_t(\phi)$ and $w = z' > 0$, the exact comparison identity

$$\frac{d}{dz} \left(\frac{w^2}{2} - \frac{2}{z} \right) = \frac{2}{z^2} \left[1 - \left(1 + \frac{r^2}{4z^2} \right)^{-3/2} \right]$$

gives $T = (\pi/4)Z^{3/2}(1 + o(1))$. Combining this with $\phi_n + 4T_n = 2\pi n$ proves the asymptotics. $\square$

Intervening regularized heavy-heavy collisions mean these restricted brake states are not classical periodic counterexamples. For finite B , they form an infinite resonance sequence at which the full velocity residual must be split.

At a fixed resonance this splitting has an exact local form. If $a_n = z_{\theta}(T_n) < 0$, $b_n = \partial_\phi z_\theta(T_n)$, and $\mathcal{A}(\phi) = \phi + 4T(\phi)$, the Jacobian of the two Jacobi radial-rate conditions with respect to phase and time is

$$D_n = Z_n(2b_n - a_n/2) = -\frac{Z_n a_n}{2} \mathcal{A}'(\phi_n).$$

Thus radial continuation by the implicit-function theorem is equivalent to simplicity of the turn-phase root. Simplicity is not required for local exclusion: uniform convergence of the divided transverse velocity makes any sequence of finite- B brakes converging to a restricted brake satisfy $0 = Z'_{B,x}/B \rightarrow C_n$. Its first coefficient is C_n/Z_n , where

$$C_n = -\frac{3}{4Z_n} \int_{-\infty}^{T_n} \frac{r^2 z p}{d^5} d\theta, \quad p'' = \frac{r^2 - 2z^2}{d^5} p, \quad p - z = O(|z|^{-1})$$

on the incoming tail. The first-order incoming theorem is now proved in smooth divided Levi-Civita normal variables. On the centered parabolic separatrix let p_- have the same incoming normalization and define

$$\mathcal{W}_\infty = \lim_{\theta \rightarrow +\infty} (z p'_- - z' p_-).$$

The exact identity

$$\frac{d}{d\theta} (z p'_- - z' p_-) = \frac{3r^2 z p_-}{2d^5}$$

makes this limit finite. Phase-uniform large-turn comparison, applied to $q = p/z$ and its Wronskian, proves

$$\boxed{Z_n C_n \longrightarrow -\mathcal{W}_\infty/2.}$$

Thus one scalar inequality $\mathcal{W}_\infty \neq 0$ implies $C_n \neq 0$ at every sufficiently late restricted resonance, uniformly over multiple roots. Ordinary finite-cutoff integrations give $\mathcal{W}_\infty \approx 0.4457$. A finite CAPD enclosure, joined to an analytic outgoing-Jost tail bound, proves

$$k_+(0) > 7/20, \quad k'_+(0) > 3/5, \quad \frac{11}{25} < \mathcal{W}_\infty = 2k_+(0)k'_+(0) < \frac{9}{20}.$$

Consequently $C_n < 0$ at every sufficiently late restricted resonance. The first-order tail theorem then proves that every fixed such resonance has a punctured finite- B neighborhood containing no full labelled brake. This fixed-resonance conclusion does not require radial simplicity. It is not uniform as $n \rightarrow \infty$, because $C_n \asymp Z_n^{-1} \rightarrow 0$, and it does not exclude a brake after later outer turns.

The joint scale is controlled by outer angular momentum. The outgoing restricted field has

$$p_- = \gamma z - \mathcal{W}_\infty \sqrt{z} + o(\sqrt{z}).$$

On the macroscopic radial Kepler arc $s = R \sin^2 \psi$, the collision-normalized transverse field $x = \sqrt{R} \sin \psi \cos \psi \sim \sqrt{s}$ solves $x'' = -2x/s^3$ and obeys $\dot{x}(\pi/2) = -1/R$. More invariantly, exact torque factorization and a collision-safe outgoing LC bootstrap prove, uniformly in first-turn height,

$$\frac{Y \times \dot{Y}}{B^{3/2}}(t_{\text{turn}}) \longrightarrow \frac{\mathcal{W}_\infty}{2} > \frac{21}{100}.$$

Hence a full labelled brake is impossible at the first outer turn throughout the accumulating skinny parabolic layer. The same tail estimate carries this positive normalized angular momentum back to a fixed large section on the inward leg. The next compact central encounter, and hence a brake after a later turn, remains uncontrolled.

One endpoint of that compact encounter map is already signed. Let (R_+, S_+) be the outgoing transverse Jost basis with $W(R_+, S_+) = -1$. Evenness and Wronskian preservation give the connection matrix

$$\begin{pmatrix} \gamma & (1 - \gamma^2)/\mathcal{W}_\infty \\ -\mathcal{W}_\infty & \gamma \end{pmatrix}.$$

The reflected returning field therefore has centered-parabolic outgoing Wronskian

$$\mathcal{C}_{2,\text{par}} = -2\gamma\mathcal{W}_\infty.$$

The extended pinned interval certificate proves $-1/100 < \gamma < -1/250$, hence $\mathcal{C}_{2,\text{par}} > 21/6250$. Thus a second-turn zero cannot emerge from the centered parabolic endpoint. The compact captured interior and its triple-collision transition remain open. On the equilateral restricted triple-collision ray the transverse coefficient is exactly $-1/(18\sigma^2)$, with indicial exponents $(3 \pm \sqrt{7})/6$; hence that endpoint requires fractional-power McGehee matching. More precisely, its remaining scalar target is

$$(\pi - \phi)^{(\sqrt{7}-1)/6} \mathcal{C}_2(\phi) \longrightarrow 4^{(\sqrt{7}-1)/6} A_- K_- > 0,$$

where A_- is the slow collision-mode coefficient of the returned Jost field and K_- is its universal first-turn Wronskian transfer.

One part of this endpoint problem is already analytic. In logarithmic time $\tau = -\log(1-s)$ and shape coordinate $y = Z/R$, the limiting base satisfies

$$y_{\tau\tau} - \frac{1}{3}y_\tau + \frac{2}{9}y[(y^2 + 1/4)^{-3/2} - 1] = 0.$$

Its increasing unstable branch from $y = -\sqrt{3}/2$ is unique modulo τ -translation, crosses $y = 0$ with $Z_s > 9^{1/3}\sqrt{5/18}$, and obeys $e^{-2\tau/3}y \rightarrow A_\infty > 0$. The last assertion follows from the strictly decreasing positive factor $e^{-2\tau/3}(y_\tau + y/3)$ and a tail contradiction. Thus the candidate universal light orbit stays separated from the prescribed binary collision. This does not yet prove that the finite-phase family selects that branch. The two scalar signs $K_- > 0$ and $A_- > 0$ are settled below.

There is also an analytic sign on the incoming half of the canonical slow transverse field. If $Q(y) = (1 - 2y^2)/(9(y^2 + 1/4)^{5/2})$ and $r = p_\tau/p$, then $r' = Q - r - r^2$ and $r \rightarrow -\alpha_-$ at the equilateral end. Tangent-line bounds for Q , together with $0 < y_\tau \leq y + \sqrt{3}/2$, make $g = -\alpha_- + (7/100)(y + \sqrt{3}/2)$ a strict lower barrier. Hence

$$p(0) > 0, \quad \frac{p_s(0)}{p(0)} \geq -\alpha_- + \frac{7\sqrt{3}}{200} > 0.$$

The passage through the prescribed binary collision and the subsequent short segment to the first turn can then be certified on a deliberately larger box. The energy identity gives $1/2 < y_\tau(0) < 3/4$, while the Riccati bound gives $p_s(0)/p(0) > 1/1000$. A pinned CAPD proof covers the center-speed interval in 256 dyadic slabs, crosses the binary collision in the analytic coordinate $\lambda = \text{sgn}(1 - s)|1 - s|^{1/3}$, and proves positive height and positive transverse derivative at the unique first turn. Therefore

$$K_- > 0.$$

The global coefficient is also positive, as proved next. The later transversality and shifted-Fuchsian argument then construct the selected local finite-phase branch.

The latter sign is nevertheless finite-dimensional. On the last binary infall, use the monotone separation r as independent variable and put $y = z/r$. The exact equations are

$$2r^2(1 - r)y_{rr} + r(3 - 4r)y_r + y[(y^2 + 1/4)^{-3/2} - 1] = 0,$$

$$2r^2(1 - r)p_{rr} - rp_r - \frac{1 - 2y^2}{2(y^2 + 1/4)^{5/2}}p = 0.$$

At equilateral collision their admissible shape exponent is $(-1 + \sqrt{19})/4$, while the transverse exponents are $\beta_\pm = (3 \pm \sqrt{7})/4$. Hence

$$\hat{A}_- = \lim_{r \downarrow 0} r^{-\beta_-} \frac{\beta_+ p - r p_r}{\beta_+ - \beta_-}$$

has the same sign as A_- . This reduces the global sign to one collision-stable/parabolic-manifold intersection and one scalar projection. A scalar comparison with the exact equilateral orbit shows that a branch with negative shape-stable coefficient remains in the cone $z < -(\sqrt{3}/2)r$ throughout the final infall. The global certificate below places every candidate parabolic connection in such a negative interval.

This match can be made square without adding a hidden dynamical parameter. On the regular section $r = 1/2$, let $U(\chi)$ be the incoming parabolic base curve, $C(a)$ the collision-stable base curve, $J(\chi)$ the returned Jost state, and $P_\pm(a)$ the canonical collision modes. The four residuals are

$$C(a) - U(\chi) = 0, \quad A_- P_-(a) + A_+ P_+(a) - J(\chi) = 0.$$

Their Jacobian is block lower triangular. In physical time the exact collision-mode Wronskian is

$$W(P_-, P_+) = -\sqrt{7},$$

so its determinant is $-\sqrt{7}$ times the oriented determinant of the two base curves. Thus only the base intersection requires a transversality certificate, after which

$$A_- = \frac{W(J, P_+)}{-\sqrt{7}}.$$

At incoming infinity, $z = -2/x^2$ and the regular fiber variables $P = x^2 p$, $\Omega = zp - \dot{z}p$ tend directly to $(2\gamma, -\mathcal{W}_\infty)$ and obey an analytic system. Hence both singular ends attach to one compact interval-Newton problem. A backward finite-cutoff diagnostic gives $a \approx -0.24695$ and $\hat{A}_- \approx 0.31 > 0$.

Theorem 16 (Global triple-endpoint coefficient). *There is exactly one collision-stable/incoming-parabolic connection in the amplitude bracket*

$$-0.24696 \leq a \leq -0.24694$$

and it satisfies $A_- > 0$.

Proof. Analytic Fuchsian fixed-point estimates attach the full collision-stable family and its canonical fast mode at $r = 2^{-32}$. A pinned interval propagation carries the whole amplitude box to a large outgoing section and proves positive velocity along every accepted step. At the left endpoint the monopole energy $H_K = w^2/2 - 2/Z$ is positive; at the right endpoint $H_K + 1/(4Z^3) < 0$. The exact tail inequality

$$0 \leq H_K(\infty) - H_K(Z) \leq \frac{1}{4Z^3}$$

and connectedness therefore supply a parabolic member, without a uniqueness or transversality assumption.

The same propagation proves $Z > r/(2\sqrt{2})$ along the whole candidate box; the analytic collision cone and the large- Z tail cover the two omitted ends. Hence $f_Z > 0$ for the scalar light equation. For two amplitudes ordered by $b = -a$, their positive position and velocity differences obey $\Delta Z'' = c(t)\Delta Z$ with $c > 0$. The velocity difference is strictly increasing, so two different amplitudes cannot both have velocity tending to zero at infinity. This proves uniqueness within the bracket and orders the turn and escape sides.

On every parabolic member, a Volterra estimate in (P, Ω) attaches the exact returned Jost data $(2\gamma, -\mathcal{W}_\infty)$ from infinity. Combining those boxes with the compact propagation gives the uniform rational margin

$$W(J, P_+) < -\frac{1}{2}.$$

Since $W(P_-, P_+) = -\sqrt{7}$, the exact projector gives

$$A_- = \frac{W(J, P_+)}{-\sqrt{7}} > 0.$$

The collision attachment is analytic; the compact enclosures use CAPD 6.1.0 at pinned commit 731079217a9254ea2948d742df2b170895effe7f. $\square$

Together with $K_- > 0$, this proves that the proposed limiting coefficient $4^{(\sqrt{7}-1)/6} A_- K_-$ is positive. It does not by itself prove the limit: one must still show that the finite-phase captured family selects this base connection and its normalized transverse field.

That singular local implication has an exact shifted-Fuchsian reduction. If ρ is the binary separation at center crossing and $\xi = -\log(r/\rho)$, then

$$2(1 - \rho e^{-\xi})y_{\xi\xi} + (-1 + 2\rho e^{-\xi})y_{\xi} + y[(y^2 + 1/4)^{-3/2} - 1] = 0.$$

The translated field converges to the universal hyperbolic saddle. A parameter-dependent inclination estimate selects its increasing unstable branch, while the fibered exchange estimate suppresses the fast mode by $\rho^{\sqrt{7}/2}$. Thus fixed-section convergence to the certified endpoint implies the full base-and-field asymptotic above. The remaining issue is the compact localization of any remotely continued captured component; the singular passage itself introduces no further sign ambiguity.

The certified connection is in fact a transverse base-curve intersection. Differentiating the collision contraction in the fixed terminal-collision gauge gives a tangent J with $J, J_s > 0$. The validated cone makes $J_{ss} = f_Z J > 0$, so its velocity component stays bounded away from zero. By contrast, the differentiable McGehee fixed-phase parabolic graph has contracting tangents, whose velocity components tend to zero at infinity. The section tangents therefore cannot be collinear.

It follows that one side of the incoming parabolic curve exits the translated saddle along the increasing universal peel. This produces a genuine local captured branch with center phase tending to π . On that branch the full fibered exchange theorem gives

$$(\pi - \phi)^{(\sqrt{7}-1)/6} \mathcal{C}_2(\phi) \longrightarrow 4^{(\sqrt{7}-1)/6} A_- K_- > 0.$$

Thus the restricted second-turn coefficient is positive on a punctured one-sided neighborhood of the certified triple endpoint. This is a collision-regularized restricted theorem. A finite-positive-mass wedge can now be transferred as follows.

Let $\omega_0(B), \omega_1(B) \rightarrow 0$ be respectively the fixed-section full-state and divided-transverse matching moduli supplied by the incoming-tail theorem. Put

$$\lambda_s = \frac{1 + \sqrt{19}}{6}, \quad \lambda_t = \frac{1 + \sqrt{7}}{6}, \quad \lambda_s > \lambda_t.$$

For an actual Pythagorean trajectory whose selected intercept $\Phi_{\text{ref}}(B)$ lies on the certified local restricted branch, let $\varepsilon = (\pi - \phi)/4$ be the comparison center-to-collision gap.

Theorem 17 (Finite-mass triple-endpoint wedge). *If $B_j \rightarrow 0$, $\varepsilon_j \rightarrow 0$, and*

$$\omega_0(B_j) + B_j = o(\varepsilon_j^{\lambda_s}),$$

then either the finite- B_j classical trajectory has a collision during the near-triple passage, or it has no full labelled brake near the continued restricted second turn. In the collision-free alternative, with $\eta_j = B_j \varepsilon_j^{-\lambda_t}$,

$$\frac{1}{\eta_j}(\mathcal{Z}_{j,x}, \mathcal{Z}_{j,x,s}) \longrightarrow -\frac{A_-}{2}(P_-, P_{-,s})$$

through the heavy-binary encounter and at the simple universal turn.

Proof. Homogeneity leaves the exact finite- B Jacobi field unchanged under $R, Z = \varepsilon^{2/3}(\mathcal{R}, \mathcal{Z})$ and $\theta = \theta_c + \varepsilon s$. If h is the signed unstable distance from the certified collision-stable graph, transversality gives $h \asymp \varepsilon^{\lambda_s}$, while the actual transverse displacement relative to the inner size is $\eta = B \varepsilon^{-\lambda_t}$.

The full homogeneous McGehee system, after zero-angular-momentum rotation reduction, has reduced log-size spectrum

$$-\frac{\sqrt{19}-1}{4}, \quad \frac{1+\sqrt{19}}{4}, \quad 1 - \frac{3-\sqrt{7}}{4}, \quad 1 - \frac{3+\sqrt{7}}{4}.$$

A simultaneous longitudinal/transverse graph-transform bootstrap bounds the translated error by

$$O\left(\frac{\omega_0 + B}{h} + B|\log h| + \eta + \omega_1\right) = o(1).$$

It also proves that the actual and restricted center scales have ratio $1 + o(1)$. At the heavy-binary encounter, reflection and Hadamard division of the smooth Levi-Civita field leave a direct source of size $B/\eta = \varepsilon^{\lambda_t} \rightarrow 0$. Thus the divided outer state passes through the regularized block and converges in physical C^1 near the separated simple turn. Since $p = -2\xi$, its limit is the one displayed. Finally $A_- > 0$ and $K_- = Z_*(s_t)P_{-,s}(s_t) > 0$, so the limiting transverse velocity is nonzero. A true finite- B collision instead terminates the classical solution. Full details, including the fixed-inertial reconstruction, are in `docs/FINITE_B_TRIPLE_ENDPOINT.md`. $\square$

In particular, with

$$\Delta(B) = (\omega_0(B) + B)^{3/(1+\sqrt{19})},$$

all sufficiently small finite masses are excluded from a brake on the local branch for $\Delta(B) \leq \varepsilon \leq \varepsilon_0$. The complementary boundary layer $0 < \varepsilon < \Delta(B)$ remains open.

There is nevertheless no brake *inside* that boundary layer's near-triple tube. Indeed the heavy pair has physical separation $X = BR$, while at every full brake energy conservation requires $U = U_0 = AB + (AB)^{-1}$. Positivity of the other pair terms gives the exact strict condition

$$|R| > \frac{A}{BU_0} = \frac{1 - B^2}{1 + B^2 - B^4} = 1 - 2B^2 + O(B^4).$$

Thus every state with $|R| = o(1)$ is separated from the brake boundary. The unresolved boundary layer concerns the signed outgoing data and possible later motion after the heavy pair re-expands, not a brake during the close passage.

Theorem 18 (Joint endpoint reduction). *Let $B_j \downarrow 0$ be collision-free finite-mass states whose selected restricted intercepts approach the certified triple endpoint. There are intrinsic longitudinal and reflection-odd unstable amplitudes H_B, T_B for which*

$$\frac{T_B}{B} \longrightarrow -\frac{\hat{A}_-}{2} \neq 0.$$

Put

$$\mu = \frac{1 + \sqrt{19}}{4}, \quad \tau = \frac{1 + \sqrt{7}}{4}, \quad \rho_T = |T_B|^{1/\tau}, \quad \kappa_B = \frac{H_B}{|T_B|^{\mu/\tau}}.$$

If $\kappa_{B_j} \rightarrow \kappa \in \mathbb{R}$, scaling size by ρ_T and time by $\rho_T^{3/2}$ gives the one-parameter planar massless restricted limit

$$\mathcal{R}_{ss} = -2\Phi(\mathcal{R}), \quad \mathcal{Z}_{ss} = -\Phi(\mathcal{Z} + \mathcal{R}/2) - \Phi(\mathcal{Z} - \mathcal{R}/2).$$

If $\kappa_{B_j} \rightarrow \pm\infty$, longitudinal scaling instead selects the corresponding rectilinear exit. A genuine finite- B collision is always the classical-termination alternative.

Proof. Use the parity-adapted nonlinear unstable amplitudes of the persisted McGehee rest point. In the auxiliary family with equal heavy masses $(1, B, 1)$, heavy exchange and reflection split the stable foliation and spectral projectors into exact even and odd blocks. Since $A(B) = \sqrt{1 - B^2} = 1 + O(B^2)$, the Pythagorean field differs from this auxiliary field by $O(B^2)$ in C^2 . Subtracting the moving stable graph therefore does not alter the divided first-order odd coefficient, and the proved Fuchsian matching gives $T_B/B \rightarrow -\hat{A}_-/2$.

After translations, $\text{SO}(2)$, log-size/time, and fixed energy have been removed, the unstable space has exactly these two scalar modes. The longitudinal stable mode, fast odd mode, and energy/clock variables lie in the stable or triangular block; no planar angle remains after zero-angular-momentum reconstruction.

The two unstable exponents satisfy $1 < \mu/\tau < 2$. Thus $H \mapsto H + O(T^2)$ changes κ_B by $o(1)$, while $B|\log B| \rightarrow 0$ absorbs finite- B exponent drift over the saddle dwell. The exponential dichotomy gives, on every fixed translated interval, the uniform error

$$O_L\left(|\kappa_B - \kappa| + B|\log |T_B|| + B + |T_B|^{2-\mu/\tau} + e^{-c|\log |T_B||}\right) + o_L(1).$$

Translation by $-\log \rho_T$ and Newtonian homogeneity therefore give the displayed one-parameter limit. Across a separated limiting heavy-binary collision the convergence is only in the common Levi-Civita variables and clock; an exact collision of a finite member is not continued. $\square$

This theorem removes the arbitrary two-parameter character of the shrinking boundary layer, but it does not yet exclude a later brake. The remaining local task is a global classification and a nonvanishing scattering theorem for the single real family indexed by κ .

Lemma 19 (Planar shape identities). *On a zero-energy radial-binary half put*

$$R = 9^{1/3}t^{2/3}e_x, \quad Z = 9^{1/3}t^{2/3}w(\zeta), \quad \zeta = \log t.$$

Then

$$w_{\zeta\zeta} + \frac{1}{3}w_{\zeta} = \nabla W(w),$$

where

$$W = \frac{1}{9} \left(|w|^2 + |w + e_x/2|^{-1} + |w - e_x/2|^{-1} \right).$$

If $\mathcal{E} = |w_{\zeta}|^2/2 - W$ and $\Lambda = w \times w_{\zeta}$, then

$$\mathcal{E}_{\zeta} = -\frac{|w_{\zeta}|^2}{3}, \quad \Lambda_{\zeta} + \frac{\Lambda}{3} = -xyK(x, y), \quad K > 0.$$

The sharp global bound is $W \geq 11/36$, with equality only at the two equilateral shapes. Hence a nonhomothetic equilateral-stable orbit has $w_{\zeta} \neq 0$ at every finite point and has no compact collision-free backward-captured component. Moreover, on $|y| \geq \sqrt{3}/2$,

$$W_{xx} \geq \frac{1}{9} \left(\frac{7}{4} - \frac{8}{3\sqrt{3}} \right) > 0.$$

Proof. The first two identities follow by direct substitution and scalar/cross products. If $d_{\pm} = |w \pm e_x/2|$, then

$$9W = \sum_{\pm} \left(\frac{d_{\pm}^2}{2} + \frac{1}{d_{\pm}} \right) - \frac{1}{4} \geq \frac{11}{4},$$

since $d^2/2 + d^{-1} - 3/2 = (d-1)^2(d+2)/(2d)$. The strict energy identity excludes a finite shape rest. A compact backward tail would have an invariant α -limit set on which the energy is constant, hence $w_{\zeta} = 0$ and $\nabla W = 0$; its energy would be at most $-11/36$, contrary to the strict stable-tail energy.

For the last inequality, set $Y = y^2$ and $g_Y(q) = (Y - 2q)/(Y + q)^{5/2}$. Then

$$9W_{xx} = 2 - g_Y((x + 1/2)^2) - g_Y((x - 1/2)^2).$$

One squared argument is at least $1/4$; monotonicity of g_Y gives an upper bound $Y^{-3/2} + (Y - 1/2)/(Y + 1/4)^{5/2}$, whose maximum for $Y \geq 3/4$ is $8/(3\sqrt{3}) + 1/4$. $\square$

The angular identity fixes a nonzero sign on every stable tail that remains in the outer strip. It is not yet global: at $\kappa = 0$ the stable manifold bends into the inner strip with the exact coefficient $(2\sqrt{21} - 5\sqrt{3})/4 > 0$. Thus the next problem is a finite transition classification, not an invariant-strip assertion. On the incoming half $t = t_c - \theta$, so the physical angular-momentum sign is opposite the ζ -oriented sign above.

For the collision shoot we use the exact quartic stable jet. With $p = Te^{-a_T\zeta}$ and $\nu = He^{-a_L\zeta}$ it has the normalized form

$$x = p + Ap\nu + Bp^3 + Cp\nu^2 + O_4, \quad q = \nu + Kp^2 + D\nu^2 + Ep^2\nu + F\nu^3 + O_4,$$

Its degree-four correction is

$$x_{[4]} = Gp^3\nu + Hp\nu^3, \quad q_{[4]} = Ip^4 + Jp^2\nu^2 + L\nu^4.$$

All twelve coefficients are explicit elements of $\mathbb{Q}(\sqrt{3}, \sqrt{7}, \sqrt{19})$. Direct substitution kills the full residual through total degree four.

Theorem 20 (Stable-tail enclosure). *Let P_4 be this quartic parameterization and equip vector power series with the weighted coefficient norm*

$$\|F\|_{r_T, r_L} = \sum_{i,j} (|F_{ij}^x| + |F_{ij}^q|) r_T^i r_L^j.$$

On $|p| \leq 1/200$, $|\nu| \leq 1/2500$, there is a unique normalized analytic stable parameterization $P = P_4 + E$, absolutely convergent on the closed polydisc, such that

$$\|E\|_{1/200, 1/2500} < \frac{1}{125000000}.$$

On the half polydisc,

$$\|E\| < \frac{1}{4000000000}, \quad \|\mathcal{D}E\| < \frac{1}{8000000000},$$

and

$$\|\partial_\kappa E\| < 10^{-9}, \quad \|\partial_\kappa \mathcal{D}E\| < 5 \cdot 10^{-9}.$$

These half-radius estimates apply to the real collision-shooting tail $p = -e^{-a_T\zeta}$, $\nu = \kappa e^{-a_L\zeta}$ for $5/4 \leq \kappa \leq 13/10$ and $\zeta \geq 10$.

Proof. The homological inverse on degrees at least two has norm below $30/17$ because its smallest divisor is $(5 + 2\sqrt{7})/18 > 17/30$. Exact algebraic coefficient bounds on P_4 and an eightyfold Cauchy dilation give residual norm $< 13/80^5$. On the base-radius ball $R = 271/50000$, a rational majorant gives $\|D^2\nabla W\| < 9$, hence contraction factor $(30/17)9R < 1/10$. The correction is consequently below $13/1671168000 < 1/125000000$. Since it begins in degree five, restriction to half radii and coefficient differentiation give the displayed state and parameter-tangent bounds. Exact Taylor lower bounds for the two exponentials put the stated real tail inside the half polydisc. $\square$

The degree-five defect can also be removed exactly. Reflection symmetry gives

$$E_5^x = Mp^5 + Np^3\nu^2 + Op\nu^4, \quad E_5^q = Pp^4\nu + Qp^2\nu^3 + R\nu^5,$$

where all six coefficients lie in $\mathbb{Q}(\sqrt{7}, \sqrt{19})$ and have signs $(+, -, +, -, +, -)$. Exact homological substitution kills every degree-five residual. Since homogeneous degrees are disjoint in the coefficient ℓ^1 norm used above, writing $P = P_4 + E_5 + E_{\geq 6}$ preserves the four norm bounds while restriction to normalized radius λ gains λ^6 instead of λ^5 . At $\zeta = 15$ the collision verifier has $\lambda < 0.04403$. The six radical expressions and exact symbolic regression are recorded in the companion quintic-enclosure note.

Lemma 21 (Forced light-collision chart). *Let $\epsilon = \pm 1$ select the heavy primary at $\epsilon R/2$ and put $q = Z - \epsilon R/2$. On either heavy-binary half let $t = |\theta - t_c|$ and $\eta = dt/d\theta = \pm 1$, where θ is physical time. Then derivatives of order two in t and θ agree, and*

$$q_{tt} = -\Phi(q) + G_\epsilon(q, t), \quad G_\epsilon = \epsilon\Phi(R) - \Phi(q + \epsilon R).$$

With complex Levi-Civita variables

$$q = u^2, \quad dt = |u|^2 d\sigma, \quad h = \frac{|qt|^2}{2} - \frac{1}{|q|},$$

the analytic regular field is

$$\begin{aligned} u_\sigma &= v, \\ v_\sigma &= \frac{h}{2}u + \frac{|u|^2 \bar{u}}{2}G_\epsilon, \\ h_\sigma &= 2\operatorname{Re}(uv\overline{G_\epsilon}), \\ t_\sigma &= |u|^2. \end{aligned}$$

It preserves $2|v|^2 - 1 - h|u|^2 = 0$ and hence $|v| = 1/\sqrt{2}$ at collision. The real-analytic chart domain is $t > 0$, $u^2 + \epsilon R(t) \neq 0$; the latter inequality excludes the other primary. At collision $t_\sigma = t_{\sigma\sigma} = 0$ and $t_{\sigma\sigma\sigma} = 1$, so the radial clock is monotone through the regularized collision. Away from collision the physical velocity is

$$Z_\theta = \eta \left(\frac{2v}{\bar{u}} + \frac{\epsilon}{2}R_t \right).$$

Thus increasing regularized time has the opposite physical orientation on the incoming half.

Consequently an isolated collision in the projective family is a zero of the square map

$$(\kappa, \sigma) \longmapsto (\operatorname{Re} u, \operatorname{Im} u).$$

Here the asymptotic normalization of κ , one LC deck lift, and the origin and orientation of σ are fixed. Moreover the selected-primary relative angular momentum is exactly

$$\ell = q \times q_t = 2\det(u, v), \quad \partial_\kappa \ell|_{u=0} = 2\det(\partial_\kappa u, v).$$

Thus the interval-Newton determinant is precisely half the derivative of a signed physical impact parameter (with the branch factor η when physical time is restored). With $dt/d\sigma = |u|^2$ fixed it is invariant under a κ -dependent LC-time translation and under a deck change; its sign still depends on the orientation of κ . If $D_c = \det(\partial_\kappa u, v) \neq 0$ at the zero, the equation $u \cdot v = 0$ defines a unique nearby local closest-passage time. Orthogonal projection onto $v_c^\perp$ and $|v_c|^2 = 1/2$ give

$$r_{\min, \text{local}} = 2D_c^2(\kappa - \kappa_c)^2 + O(|\kappa - \kappa_c|^3), \quad \ell = 2D_c(\kappa - \kappa_c) + O(|\kappa - \kappa_c|^2).$$

This proves local avoidance on both punctured sides of a transverse zero, without asserting absence of other collisions. This was the interval-Newton target closed below. The resulting transverse zero separates nearby continuations locally; identifying full adjacent collision-free components still requires excluding every remaining collision zero. Its regularized continuation is not a physical continuation of a classical finite-mass collision.

Validated finite stage. For $1.26793 \leq \kappa \leq 1.26794$, pinned CAPD 6.1.0 native-interval integration propagates the rigorous stable-tail box from $\zeta = 10$ to $\zeta = 0.3$, converts it to the selected-primary LC chart, and propagates the reversed LC flow for the exact duration 1071249/1000000. Every accepted LC step satisfies $t > 1$, $R > 2$, and distance to the other primary greater than one. At the final fixed time,

$$-0.017807 < \operatorname{Re} u < 0.016661, \quad -0.003953 < \operatorname{Im} u < 0.004019.$$

This validates finite propagation and chart safety, but not a collision zero: the present interval coordinate conversion loses the affine κ correlation, so the terminal box is too wide for interval Newton.

Theorem 22 (Transverse planar collision boundary). *The normalized lower-equilateral planar family with $T = -1$ has exactly one selected-primary collision zero in*

$$\begin{aligned} 1.2679350755 &\leq \kappa \leq 1.2679352755, \\ 1.0712484057 &\leq -\sigma \leq 1.0712486057. \end{aligned}$$

It lies in

$$1.2679351752 < \kappa_c < 1.2679351759, \quad -1.0712485063 < \sigma_c < -1.0712485051,$$

and is transverse, with

$$-3.52212 < \det(\partial_\kappa u, v) < -3.38401.$$

Proof. At $\zeta = 15$, restriction of the analytic tail enclosure gains the factor λ^5 , where

$$\lambda = \max\{400e^{-15a_T}, 6500e^{-15a_L}\} < 0.04403,$$

because the state and parameter-tangent corrections all begin in total degree five. A pinned CAPD C^1 doubleton is retained through three fields: reversed shape flow to $\zeta = 0.3$, a unit construction flow writing the exact nonlinear LC lift, and reversed LC flow. Whole-step enclosures keep both shape primary distances positive, prove $y > 0$ and $u_r, u_i > 0$ at the selected lift, and retain $t > 1$, $R > 2$, and other-primary distance greater than one.

For $S = -\sigma$, the computed Jacobian columns are $(\partial_\kappa u, -v)$. Its determinant lies in

$$[3.3840110561, 3.5221124176].$$

Rigorous interval Gaussian elimination puts the interval Newton image strictly inside the displayed parameter-time box, proving existence and uniqueness there. Restoring increasing σ gives the stated determinant sign. The hexadecimal certificate and pinned replay are recorded in the companion computer-assisted proof note. $\square$

The adjacent planar component contains a second collision boundary, involving the opposite primary.

Theorem 23 (Second planar collision boundary). *The same normalized planar family has exactly one opposite-primary collision zero in*

$$\begin{aligned} 1.2640119181 &\leq \kappa \leq 1.2640119321, \\ 1.0275748704 &\leq S_2 \leq 1.0275749704. \end{aligned}$$

It lies in

$$1.2640119191 < \kappa_2 < 1.2640119311, \quad 1.0275748796 < S_2 < 1.0275749612,$$

is reached without an earlier collision, and is transverse. For the fixed second-chart clock, the interval Jacobian determinant lies in

$$98.8377 < \det D_{(\kappa, S_2)} F_2 < 99.2930.$$

Proof. Propagate the stable-tail enclosure in the positive-primary LC chart for the exact duration $7/4$. Whole-step intervals give $t > 1$, $R > 2$, distance to the opposite primary greater than one, and selected-primary distance greater than 10^{-8} . Thus this leg is classically collision-free. At its fixed endpoint make the exact chart change

$$q_- = q + Re_x, \quad \dot{q}_- = \frac{2uv}{|u|^2} + \frac{2R}{3t}e_x, \quad u_-^2 = q_-.$$

The switch enclosure proves $\text{Im } q_- < 0$, and its image fixes the exact lift by $\text{Re } u_- > 1$, $\text{Im } u_- < 0$. In the resulting forced LC chart the positive primary stays at squared distance greater than $1/100$, $t > 1/2$, and $\text{Re } v_- > 1/2$ through the candidate collision. Since $(\text{Re } u_-)_S = -\text{Re } v_- < 0$ and $\text{Re } u_- > 1$ at entry, the terminal zero is the first selected- pair collision rather than a regularized recollision. A second C^1 doubleton therefore encloses the square residual $F_2 = (\text{Re } u_-, \text{Im } u_-)$. Pinned interval Newton maps the displayed box strictly into

$$[1.264011919130, 1.264011931053] \times [1.027574879669, 1.027574961151].$$

The displayed determinant interval excludes zero, proving existence, uniqueness in the box, and transversality. The collision time satisfies $0.6152 < t < 0.6153$, so it precedes the prescribed heavy-binary collision. The hexadecimal certificate is recorded in the second-collision proof note. $\square$

Theorem 24 (Second-boundary collision-or-escape box). *For every limiting planar parameter*

$$1.2640099161 \leq \kappa \leq 1.2640126461,$$

the classical branch either collides or, if it remains collision-free, reaches a strict hyperbolic-escape cone. The regularized continuation of the exact collision root reaches that cone. Every sufficiently small compatible positive-light-mass member near this box likewise either collides classically or escapes, and hence has no second brake.

Proof. Continue the negative-primary LC box beyond the preceding root enclosure to a fixed separated section and set

$$q_+ = q_- - Re_x, \quad \dot{q}_+ = \frac{2u_-v_-}{|u_-|^2} - \frac{2R}{3t}e_x.$$

The interval image proves $\text{Im } q_+ < 0$. On that sheet use the cancellation-free exact lift

$$u_{+,i} = -\sqrt{\frac{|q_+| - q_{+,x}}{2}}, \quad u_{+,r} = \frac{q_{+,y}}{2u_{+,i}}, \quad v_+ = \frac{\overline{u_+}\dot{q}_+}{2}.$$

The resulting forced positive-primary LC field has nonsingular forcing

$$g_+ = \frac{e_x}{R^2} - \frac{q_+ + Re_x}{|q_+ + Re_x|^3}.$$

Pinned CAPD carries 195 abutting root-width tiles through this third chart and then through the signed- $\lambda = \sqrt[3]{t}$ heavy-binary bridge. The tile centers are $1.2640119251 + 14j \cdot 10^{-9}$, $-143 \leq j \leq 51$, and every radius is 7×10^{-9} ; their outward-rounded endpoints cover the displayed interval. The squared distances to the unregularized primary in the two post-root LC blocks, and to either light primary in the bridge, are respectively greater than

$$0.1760, \quad 0.1264, \quad 0.004936.$$

At the terminal section the restricted escape inequality has margin greater than 2.7591, while the hierarchical finite-mass inequality has margin greater than 2.8882. Therefore any selected-pair zero is a terminating classical collision; every member without such a zero agrees with the regularized flow and escapes.

All preterminal maps form a finite analytic chain of pair-LC charts and a heavy-pair LC chart, with every nonselected denominator uniformly separated. Thus the Newtonian-scaled terminal state and inner energy converge for small positive light mass. The branch-independent hierarchical-energy lemma absorbs the full future $O(B)$ tidal work, transferring the strict second margin. This proves the positive-mass dichotomy without physically continuing any collision. The hexadecimal enclosure and trusted computing base are recorded in the companion second-boundary escape note. $\square$

Theorem 25 (Third and fourth planar collision boundaries). *Immediately below the preceding collision-or-escape box, the limiting planar family has two further classically reached transverse collision roots. The positive-primary root satisfies*

$$\begin{aligned} 1.2640090993171269 &\leq \kappa_3 \leq 1.2640090999643190, \\ 0.5884119895288813 &\leq S_3 \leq 0.5884197454438135, \end{aligned}$$

and

$$5224.5883 < \det D_{(\kappa, S_3)} F_3 < 5227.2175.$$

The subsequent negative-primary root satisfies

$$\begin{aligned} 1.2640090985775962 &\leq \kappa_4 \leq 1.2640090992806481, \\ 0.3710663877519410 &\leq S_4 \leq 0.3726139177003617, \end{aligned}$$

and

$$417669.25 < \det D_{(\kappa, S_4)} F_4 < 429486.31.$$

The exact member at the fourth root avoids the preceding positive-primary collision: outward-rounded interval subtraction gives a separation greater than 1.2058×10^{-11} between the corresponding root projections. The ordinal labels refer to the successive encounter charts; uniqueness is claimed only in the displayed boxes.

Proof. Continue the negative-primary LC chart from the common incoming section and make the exact positive-primary switch used in the preceding theorem. In reversed regularized time, put

$$F_3(\kappa, S_3) = (\operatorname{Re} u_+, \operatorname{Im} u_+), \quad \partial_{S_3} u_+ = -v_+.$$

Pinned interval Newton maps the input box

$$[1.2640090992, 1.2640091000] \times [0.588396, 0.588436]$$

strictly into the first displayed enclosure. Every earlier selected distance is positive, the nonselected primary stays separated, and $\text{Im } v_+ < -1/2$ throughout the positive-primary chart. Hence $\text{Im } u_+$ increases strictly and the enclosed zero is the first selected collision.

For the next passage reconstruct

$$\lambda = t^{1/3}, \quad R = 9^{1/3}\lambda^2, \quad Z = q_+ + \frac{R}{2}e_x, \quad V = \dot{q}_+ + \frac{R}{3t}e_x,$$

propagate the signed- λ bridge for the exact duration 97/5000, and set

$$q_- = Z + \frac{R}{2}e_x, \quad \dot{q}_- = V + \frac{9^{1/3}}{3\lambda}e_x.$$

On the certified sheet $q_{-,x} > 0.05$, the lift

$$u_{-,r} = \sqrt{\frac{|q_-| + q_{-,x}}{2}}, \quad u_{-,i} = \frac{q_{-,y}}{2u_{-,r}}, \quad v_- = \frac{\overline{u_-}\dot{q}_-}{2}$$

is analytic even across $q_{-,y} = 0$. With $F_4 = (\text{Re } u_-, \text{Im } u_-)$, interval Newton maps

$$[1.2640090985, 1.26400909929] \times [0.37084, 0.37284]$$

strictly into the second displayed enclosure. The other primary remains at squared distance greater than 1/10, and $\text{Re } v_- > 1/2$ makes $\text{Re } u_-$ decrease strictly to its first zero.

Finally, a separate interval-Newton replay on

$$[1.2640090981, 1.2640091001] \times [0.588316, 0.588516]$$

contains every possible intervening positive-primary zero and proves that its unique root has

$$1.2640090992927064 \leq \kappa_3 \leq 1.2640090999346494.$$

Subtracting the fourth-root projection with outward rounding gives the stated positive gap. The positive chart stays past its zero sheet thereafter, while all bridge and fourth-chart separations are strict. Thus the fourth root is reached classically without the third collision. The pinned hexadecimal enclosures and complete chart-domain audit appear in the companion proof note. $\square$

Theorem 26 (Outgoing entry after the fourth boundary). *For every*

$$1.2640090985 \leq \kappa \leq 1.26400909929,$$

the limiting classical branch either collides or reaches a uniformly analytic fifth positive-primary LC chart. Along the regularized common continuation, the intervening negative-chart other-primary squared distance is greater than 0.333865, while at the switch

$$|u_-|^2 > 0.213057, \quad q_{+,x} < -0.577809, \quad |q_+|^2 > 0.333864.$$

No terminal escape assertion is made.

Proof. Evaluate the fourth negative-primary chart at the fixed reversed clock $S_4 = 0.37284$, which is strictly beyond the preceding root-clock enclosure, and continue for the exact additional duration

0.42716. Pinned whole-step enclosures give the displayed separation bounds and keep $t > 0.05$. The exact switch

$$q_+ = u_-^2 - Re_x, \quad \dot{q}_+ = \frac{2u_-v_-}{|u_-|^2} - \frac{2R}{3t}e_x$$

is therefore nonsingular. Since $q_{+,x} < -0.05$, the sheet

$$u_{+,i} = -\sqrt{\frac{|q_+| - q_{+,x}}{2}}, \quad u_{+,r} = \frac{q_{+,y}}{2u_{+,i}}$$

is uniformly analytic; the interval image has $u_{+,i} < -0.760137$ and $t > 0.240640$. A selected-pair zero anywhere in the chain is a terminating classical collision. Conditional on no such zero, classical uniqueness identifies the trajectory with the regularized chain and proves arrival at the fifth chart. The pinned hexadecimal record appears in the companion proof note. $\square$

Theorem 27 (Outgoing section after the fifth passage). *For every*

$$1.2640090985 \leq \kappa \leq 1.26400909929,$$

the limiting classical branch either collides or crosses the outgoing fifth positive-primary LC section $c_i = 1/20$. At the crossing,

$$|c|^2 > 0.002166715533586436, \quad r_{\text{other}}^2 > 0.4599473198119462, \quad -d_i > 0.740153775124572.$$

In particular the section is collision-free and uniformly transverse. No terminal escape assertion is made.

Proof. Use the exact quintic tail enclosure at $\zeta = 15$ and retain one CAPD doubleton from the stable tail. Dimension-preserving nonlinear chart maps act directly on this set. For a dimension-changing map $y = F(x)$, append dormant coordinates and integrate the exact auxiliary graph system

$$x' = 0, \quad y' = F(x), \quad y(0) = 0$$

for one auxiliary unit. This preserves all correlation generators through dimensions $13 \rightarrow 19 \rightarrow 24 \rightarrow 30$ instead of replacing each nonlinear image by an interval product. The final 30-dimensional system freezes the old coordinates and propagates the fifth six-dimensional LC state to the Poincare section. The pinned interval image gives the three displayed strict bounds. Every ordinary denominator and square-root sheet in the preceding chain stays inside its analytic domain. A selected-pair zero is a terminating classical collision; conditional on no such zero, uniqueness identifies the regularized and classical branches up to the section. Auxiliary graph time is bookkeeping and is not physical flight time. $\square$

The next two close passages overlap, so a single-centre chart is insufficient. They admit one simultaneous polynomial regularization. Write

$$s = \sin(\xi/2), \quad c = \cos(\xi/2), \quad s^2 + c^2 = 1, \quad z = \frac{c^2 - s^2}{2}, \quad A = -sc, \quad g = |A|^2.$$

Then the two primary distances are $|s|^2$ and $|c|^2$, so g is their product. If

$$E = \frac{|z_\zeta|^2}{2} - W(z), \quad p = g\xi_\zeta, \quad \frac{d\zeta}{d\sigma} = \eta g,$$

put

$$\begin{aligned} K_0 &= \frac{2}{9}Az(\bar{A}^2 - \bar{z}^2), \\ K_{\text{pair}} &= \frac{s\bar{c} - c\bar{s}}{9}, \\ \mathcal{P} &= -2A\bar{z}E + K_0 + K_{\text{pair}} - \frac{g}{3}p. \end{aligned}$$

The simultaneous two-centre field is

$$s_\sigma = \eta cp/2, \quad c_\sigma = -\eta sp/2, \quad p_\sigma = \eta \mathcal{P}, \quad E_\sigma = -\eta |p|^2/3, \quad \zeta_\sigma = \eta g.$$

It is polynomial in real and imaginary parts. It preserves $s^2 + c^2 = 1$ and

$$|p|^2 - 2gE - \frac{2}{9}(g|z|^2 + |s|^2 + |c|^2) = 0.$$

At either collision the latter gives $|p|^2 = 2/9$. From the fifth positive LC state $q_+ = u^2$ it is initialized algebraically by $s = iu/\sqrt{R}$, $c = \sqrt{1 - s^2}$. Thus the remaining obstruction is a finite validated propagation in a regular field, not the existence of a common analytic chart.

Theorem 28 (Fourth-boundary collision–ejection classification). *The fourth limiting collision root has the refined enclosure*

$$\begin{aligned} 1.26400909893331527 &\leq \kappa_4 \leq 1.26400909893331813, \\ 0.371840180067887265 &\leq \tau_4 \leq 0.371840186320052546, \end{aligned}$$

and its transverse negative-primary LC velocity satisfies

$$-3.832349985449157 \cdot 10^{-7} \leq v_i \leq -3.681944345249734 \cdot 10^{-7} < 0. \quad (38)$$

On the regularized collision–ejection continuation, the following positive and negative two-centre foci are both noncollisional. Their normalized selected distances obey

$$r_+ \geq 3.964960196340643 \cdot 10^{-11}, \quad r_- \geq 1.098165821369775 \cdot 10^{-6}. \quad (39)$$

At the next positive focus the selected distance exceeds 0.0506366673765623. The continuation then reaches the strict restricted escape cone with margin greater than 10.2599662232953, without another light–primary collision; the finite-mass-form comparison expression has margin greater than 10.3953509156741.

Proof. The degree-eight stable graph leaves a rigorously bounded order-nine tail at the entrance section. A 200-bit MPFR-backed CAPD integration and interval Newton step first maps the fourth-root box strictly into

$$[1.26400909893313121, 1.26400909893350310]$$

in the parameter coordinate. A nested Newton localization of the same unique root gives the displayed parameter and clock boxes. Re-evaluation on that image gives (38). The nesting is only a refinement: existence and uniqueness come from the first strict inclusion.

At the negative collision introduce elliptic two-centre coordinates

$$\alpha = \pi, \quad \beta = 0, \quad p_r = -\frac{2v_r}{3}, \quad p_i = -\frac{2v_i}{3}.$$

The first positive focus is $\alpha = 2\pi$, where $r_+ = \sinh^2(\beta/2)$. An outward-widened rational hull of the independent multiprecision certificate propagates under the simultaneous regularized field to

$$-1.310804394749059 \cdot 10^{-5} \leq \beta \leq -1.259358598132686 \cdot 10^{-5},$$

which proves the first inequality in (39). The next negative focus has

$$-0.002181489558789266 \leq \beta \leq -0.002095867762413745,$$

and hence the second inequality in (39).

At the next positive focus, embed the exact two-centre-to-ordinary map as an auxiliary graph ODE instead of taking a componentwise interval hull. This retains all six incoming doubleton generators. The resulting ordinary heavy-binary bridge crosses the regularized binary section and reaches the terminal escape section with both light–primary squared distances greater than 0.00262385439850217. Direct outward-rounded evaluation gives the two strict escape margins stated above. The pinned source versions, hexadecimal input boxes, and reproduction commands are recorded in the companion computer-assisted proof note. $\square$

The theorem classifies the mathematical collision–ejection continuation of the boundary member. Its classical Newtonian solution itself ends at the fourth collision; no postcollision trajectory is being declared physical.

Corollary 29 (Local limiting collision or escape). *There is an open parameter interval containing κ_4 such that every member of the limiting planar family either suffers a terminating classical collision or, if it remains collision-free, enters the strict escape cone. In particular no collision-free member in this interval is periodic.*

Proof. Take the finite regularized chart chain from a compact incoming section to the terminal escape section. All nonselected distances and all chart sheets have strict margins on the certified boundary chain. The fourth collision is transverse by (38), so the regularized flow and the collision sheet depend analytically on the initial parameter near the root. Compactness gives a single neighborhood on which the chart chain is defined until either a selected collision occurs or the terminal section is reached. In the latter case the strict clearance and escape inequalities persist by continuity. The former case terminates the classical orbit, while the latter excludes a second brake and every bounded periodic orbit. $\square$

Theorem 30 (Explicit common-clock continuum). *Every member of the limiting planar family with*

$$1.264009099014 \leq \kappa \leq 1.264009099457 \tag{40}$$

either has a terminating classical collision before the certified terminal section, or its collision-regularized continuation reaches the strict restricted and finite-mass-form escape cones. Hence no collision-free member of (40) is periodic. This is an explicit continuum adjacent to the fourth collision boundary, not merely a certificate for one boundary member.

Proof. Let $[\kappa_4]$ and $[\tau_4]$ be the two MPFR enclosures in the fourth-boundary theorem and set $\bar{\tau} = 37184019/10^8 > \sup[\tau_4]$. Propagate the rational hull of the certified collision state for $\bar{\tau} - [\tau_4]$, obtaining a reference state X_* at a fixed fourth-LC clock. For each parameter tile K , a persistent C^1 propagation on the hull $H = \text{hull}(K, [\kappa_4])$ gives the rigorous mean-value enclosure

$$X(K, \bar{\tau}) \subset X_* + \partial_\kappa X(H, \bar{\tau})(K - [\kappa_4]). \tag{41}$$

Thus no componentwise projection of an implicitly timed fifth-section image is used.

For the negative-primary LC variables $q = U^2, V, h, t$, put

$$R = 9^{1/3}t^{2/3}, \quad C = U/\sqrt{R}, \quad S = \sqrt{1 - C^2}.$$

The exact entry to the simultaneous two-centre chart is regular at $U = 0$:

$$\begin{aligned} p &= -\frac{2}{3}\bar{S}V + \frac{2}{9t}\bar{S}|U|^2U, \\ E &= \frac{Rh}{9} - \frac{4t \operatorname{Re}(U\bar{V})}{3R^2} + \frac{2|U|^4}{9R^2} - \frac{1}{9} \left(\left| \frac{U^2}{R} - \frac{1}{2} \right|^2 + \frac{1}{|S|^2} \right). \end{aligned} \quad (42)$$

At collision these reduce to $p = -2V/3$ and $E = Rh/9 - 5/36$. Direct symbolic substitution using $R^{3/2} = 3t$ verifies (42).

From (41)–(42), a directed Poincare map reaches the first section $\alpha = 2\pi$ with $\beta < 0$. Whole-step enclosures prove $p_r < 0$, so $\alpha_\sigma = -p_r > 0$. After reboxing this six-dimensional image, a second directed map reaches $\beta = -1$ with $p_r < 0 < p_i$ throughout and $\alpha < 3\pi$. Hence no intervening focus is missed. The exact bridge map keeps both light–primary squared distances above 10^{-3} while the signed heavy coordinate crosses zero, and its regularized continuation satisfies both terminal escape inequalities.

Finally use the tiles

$$\kappa(p, r) = \frac{1264009098895 + p}{10^{12}} + [-r, r]10^{-12}$$

with $p = 120, 122, \dots, 198$, $r = 1$, and $p = 200, 204, \dots, 560$, $r = 2$. The 40 and 91 tiles overlap, and their union is exactly (40). Every tile passes the directed-section, whole-step, bridge, and terminal strict inequalities. An earlier zero is a classical collision; conditional on no such zero, uniqueness identifies the regularized and classical arcs up to the isolated heavy collision. This proves the stated dichotomy without any sampling inference.

The aggregate certificate gives, conservatively, first-focus $\beta < -0.0132$, selected squared distance $> 4.4 \times 10^{-5}$, remaining 3π -focus gap > 0.285 , bridge primary-distance square > 0.019 , restricted escape margin > 0.804 , and finite-mass-form margin > 0.935 .

The current interval-Newton projection for the third collision root overlaps (40) but extends beyond its upper endpoint. Consequently this theorem makes no claim that (40) crosses that exact root. $\square$

Theorem 31 (Positive-mass fourth-boundary cusp). *There are a constrained incoming-section neighborhood of the fourth limiting root and $B_0 > 0$ such that every compatible member with $0 < B < B_0$ either suffers a finite classical collision or its light body escapes to infinity. Consequently every collision-free member of this local cusp is nonperiodic.*

Proof. The fourth root selects the primary of mass A . In the finite-mass Jacobi variables put

$$q = Z + \frac{R}{1+A}, \quad Q = R - \frac{B}{A+B}q.$$

Exact subtraction of the Jacobi equations gives

$$q_{\theta\theta} = -(A+B)\Phi(q) - \Phi(R) - \Phi(q-R)$$

and

$$Q_{\theta\theta} = -\frac{1+A+B}{A+B} \left[A\Phi \left(Q + \frac{B}{A+B}q \right) + B\Phi \left(Q - \frac{A}{A+B}q \right) \right].$$

Thus $q = u^2$, $dt = |u|^2 d\sigma$ gives a real-analytic collision block at $u = 0$, with preserved constraint

$$2|v|^2 - (A+B) - h|u|^2 = 0.$$

The other-primary separation is strict, so the fourth transverse collision persists on a smooth local sheet.

On the limiting collision–ejection branch, a collision in elliptic two-centre coordinates can occur only at an integral- π focus with $\beta = 0$. The validated replay proves $p_r < 0$, and hence $\alpha_\sigma = -p_r > 0$, on the whole-step enclosures through the next three foci. The strict distances in (39), together with the seventh-focus and bridge clearances, therefore make the segment from a fixed fourth-LC ejection section to a fixed heavy-pair LC entry compact and collision-free. Ordinary continuous dependence transfers that segment. The collision-adapted heavy-pair chart transfers the prescribed binary crossing and the scaled terminal state.

Finally apply the branch-independent hierarchical Jacobi lemma with $\epsilon = 1/100$ and $c = 3/2$. Its limiting terminal margin is greater than 10.3953509156741. The scaled inner energy tends to zero, the future tidal integral is uniformly bounded, and its coefficient is the light mass B . Thus the strict comparison persists for small B . A true collision is classical termination; conditional on no collision, the light body escapes. $\square$

The first root’s transversality has an exact positive-mass consequence. In the full late Jacobi equations put $A = \sqrt{1-B^2}$, $M = 1+A$, and select the same primary by

$$q = Z - \frac{A}{M}R.$$

Direct subtraction gives

$$q_{\theta\theta} = -(1+B)\Phi(q) + A\{\Phi(R) - \Phi(q+R)\}.$$

The R equation still contains $B\Phi(q)$, so the common analytic collision chart uses the colliding-pair center coordinate

$$Q = R + \frac{B}{1+B}q.$$

Then

$$Q_{\theta\theta} = -\frac{M+B}{1+B} \left[\Phi \left(Q - \frac{B}{1+B}q \right) + B\Phi \left(Q + \frac{1}{1+B}q \right) \right].$$

This contains no singular pair force. After $q = u^2$ and $dt = |u|^2 d\sigma$, the equations for (Q, Q_t) acquire an exterior $|u|^2$ and are analytic at $u = 0$ when $Q \neq 0$. Hence the finite-mass LC constraint is

$$2|v|^2 - (1+B) - h|u|^2 = 0,$$

and $|v|^2 = (1+B)/2$ at collision. At $B = 0$, $Q = R$; the validated inequalities $R > 2$ and other-primary distance greater than one therefore persist on a sufficiently small common chart.

Corollary 32 (Persistence of the collision boundary). *Fix a regular constrained incoming section before the selected LC block and extend the limiting coordinate κ to a local section coordinate $\hat{\kappa}$. The certified limiting collision continues uniquely under all sufficiently small positive-mass and*

compatible section-state perturbations δ . The continued root $(\widehat{\kappa}_c(B, \delta), \sigma_c(B, \delta))$ is transverse. At fixed (B, δ) its local closest-passage normal form is

$$r_{\min, \text{local}} = \frac{2D_{B, \delta}^2}{1+B} (\widehat{\kappa} - \widehat{\kappa}_c(B, \delta))^2 + O(|\widehat{\kappa} - \widehat{\kappa}_c(B, \delta)|^3),$$

where $D_{B, \delta} = \det(\partial_{\widehat{\kappa}} u, v)$ remains negative.

Proof. The constraint, gauge, and section functions have constant full rank at the certified collision-free base state, so the fixed constrained section is smooth also at $B = 0$; equivalently use the analytic signed- B extension and then restrict to $B \geq 0$. On this section choose a C^1 chart $X_{\text{in}}(B, \widehat{\kappa}, \delta)$ which restricts at $(B, \delta) = 0$ to the validated limiting curve. Ordinary pre-LC flow, the strict square-root lift, and the collision-adapted (u, Q) flow make the square shooting map C^1 in $(B, \widehat{\kappa}, \delta, \sigma)$. The parameter-dependent implicit-function theorem and the strict determinant bound give existence, local uniqueness, and transversality. Orthogonal projection onto $v^\perp$, followed by $q = u^2$ and $|v|^2 = (1+B)/2$, gives the displayed coefficient. $\square$

Theorem 33 (Differentiated joint inclination). *Let K be a compact projective-amplitude interval inside the validated pre-collision chart and set $p = \mu/\tau$. In the normalized family $H = \kappa|T|^p$, with stable entrance coordinate S fixed, translate the logarithmic saddle dwell by $L_T = -\tau^{-1} \log |T|$ and scale size by $\rho_T = |T|^{1/\tau}$. If*

$$B|\log |T|| + B + |T|^{2-p} + e^{-cL_T} + \|\widehat{\Xi}\| \longrightarrow 0,$$

where $\widehat{\Xi}$ contains the Newtonian-scaled energy, clock, and remaining triangular deviations, then the state and its κ -tangent converge on every common collision-free compact interval. Through the selected limiting collision the same assertion holds in the scaled (u, v, Q, P, h) Levi-Civita chart.

Consequently the certified collision root extends uniquely as a C^1 graph

$$\kappa = \kappa_c(B, T, S, \Xi), \quad \kappa_c(B, T, S, \Xi) \longrightarrow \kappa_c.$$

Its determinant retains the certified negative sign. At fixed parameters,

$$r_{\min, \text{local}} = \rho_T \frac{2\widehat{D}^2}{1+B} (\kappa - \kappa_c)^2 + o(\rho_T |\kappa - \kappa_c|^2).$$

Proof. Choose a fixed weight $\mu/2 < \omega < \tau$. For small B it also satisfies $\mu_B/2 < \omega < \tau_B$, and the quadratic Lyapunov–Perron Green denominator is bounded below by $2\omega - \mu_B > 0$. Construct only the nonlinear unstable-manifold shadow on the infinite weighted tail. Translation changes its two homogeneous coefficients by $e^{(\mu_B - \mu)L_T}$ and $e^{(\tau_B - \tau)L_T}$; these tend to one because $BL_T \rightarrow 0$. Differentiating the uniformly contracting fixed-point equation gives convergence of the κ -tangent.

A bounded finite-entrance stable datum does not belong to that half-line weighted ball. Instead, C^1 stable-foliation holonomy compares the actual orbit and tangent with the shadow at one fixed incoming-tail section, with error $O(e^{-c(L_T - A)} + \rho_T |E|) + o(1)$. Compact variational flow and the zero-angular-momentum fixed-axis reconstruction propagate this estimate. The scaled collision-adapted Q -LC vector field is analytic at $u = 0$ and propagates the variational equation through the regularized zero. The parameter-dependent implicit-function theorem and the certified determinant give the graph. The C^1 expansion of u transverse to the root, together with $q = u^2$ and $|v|^2 = (1+B)/2$, gives the last formula. $\square$

Theorem 34 (Real collision sampling). *There are infinitely many distinct real parameters on the tied normalized Pythagorean curve, accumulating at the skinny endpoint, whose classical solution has a finite-time binary or triple collision.*

Proof. Let χ_* be the asymptotic intercept of the certified transverse intersection between the restricted incoming parabolic curve and the collision-stable curve. In the signed longitudinal amplitude H_0 , choose two compact intercept arcs on opposite sides of χ_* with fixed strict opposite signs. Since

$$\Phi'_{\text{ref}}(B) = -\frac{3\pi}{2}B^{-5/2}(1 + o(1)),$$

every sufficiently late phase winding supplies a disjoint real B -bracket whose reference intercept crosses these arcs. If any member collides before the selected late chart, that bracket already supplies the theorem. If not, uniform incoming-tail convergence and compact regularized flow transfer the two H -signs to the positive-mass saddle entrance.

Put $p = \mu/\tau$. Continuity gives, inside the bracket, a last zero of H followed by a first point at which $H = B^{p/2}$. On the intervening segment $0 \leq H \leq B^{p/2} \rightarrow 0$. Local uniqueness of the transverse restricted intersection therefore forces every sequence in this segment to approach the certified endpoint. Uniformly there,

$$\frac{T_B}{B} \rightarrow \Theta_* \neq 0, \quad \|\widehat{\Xi}_B\| \rightarrow 0,$$

so the normalized collision graph is defined. The continuous function

$$G(B) = H_B - |T_B|^p \kappa_c(B, T_B, S_B, \Xi_B)$$

is negative at $H = 0$ because $\kappa_c > 0$, and positive at $H = B^{p/2}$ because $|T_B|^p = O(B^p) = o(B^{p/2})$. Its intermediate-value zero is an exact selected light-heavy collision of the tied positive-mass trajectory. The phase brackets are pairwise disjoint and tend to zero. $\square$

Theorem 35 (Explicit local collision-or-escape cusp). *Every regularized limiting member in*

$$1.2679251755 \leq \kappa \leq 1.2679451755$$

reaches the strict terminal escape cone described below. There is a neighborhood of this closed interval such that every sufficiently small compatible positive-light-mass member either has a finite classical collision before the terminal section or its light body escapes to infinity. Hence every collision-free member in this local cusp has no second brake.

Proof. Ten adjacent pinned interval-boxed propagations cover the full displayed κ -interval, cross any selected collision in its Levi-Civita chart, and reconstruct the restricted outer state at a fixed ejection section. With $\lambda = \sqrt[3]{t}$, a second analytic LC bridge crosses the prescribed heavy-binary collision. Whole-step enclosures keep both light-primary squared distances above 0.89. On the outgoing section they prove the strict restricted escape inequality

$$\dot{\rho}_0 - \frac{2}{2w_0} - 2 - a'(s_0) > 0.54.$$

For the positive-mass transfer, let x be the heavy-pair Jacobi vector, y the light position relative to its center of mass, and put

$$r = |x|, \quad \rho = |y|, \quad w = \rho - r, \quad e = \frac{1}{2}|\dot{x}|^2 - \frac{M}{r}, \quad M = m_a + m_b, \quad \mathcal{M} = M + m_c.$$

The exact Jacobi equations give

$$|\ddot{y}| \leq \frac{\mathcal{M}}{w^2}, \quad |\dot{e}| \leq \frac{2m_c r |\dot{x}|}{w^3}.$$

Fix $\epsilon, c > 0$, set

$$v_b = \sqrt{\frac{2M}{r_0}} + 2\epsilon, \quad K = \int_0^\infty \frac{\sqrt{2M(r_0 + v_b t) + 2\epsilon(r_0 + v_b t)^2}}{(w_0 + ct)^3} dt.$$

If

$$\dot{\rho}_0 - \frac{\mathcal{M}}{cw_0} > v_b + c, \quad e_0 + 2m_c K < \epsilon,$$

then a simultaneous bootstrap gives

$$r(t) \leq r_0 + v_b t, \quad \dot{\rho}(t) > v_b + c, \quad w(t) \geq w_0 + ct, \quad e(t) < \epsilon.$$

Thus only an inner collision can interrupt outer escape. In the certified massless terminal state $M = \mathcal{M} = 2$, $m_c = 0$, and $e_0 = 0$. For $\epsilon = 1/100$, $c = 3/2$, interval propagation proves the stronger full-pair margin

$$\dot{\rho}_0 - \frac{2}{(3/2)(\rho_0 - r_0)} - \sqrt{\frac{4}{r_0} + \frac{2}{100}} - \frac{3}{2} > 0.65.$$

Apply the comparison in the Newtonian-scaled finite-mass terminal coordinates, where $\hat{e} = \rho_T e$. Scaled LC convergence gives $\hat{e}_0 \rightarrow 0$, while the correspondingly scaled K is uniformly bounded and $m_c K = O(B)$. The strict inequalities persist. An exact positive-mass collision terminates the classical branch; otherwise the member reaches the terminal section and escapes. $\square$

Corollary 36 (Open-window refinement). *Every sufficiently late skinny phase-cycle bracket either contains a classical collision before the common selected chart, or contains a nonempty open tied parameter interval every member of which collides or escapes.*

Proof. On a bracket without an earlier collision, use the segment in the real collision-sampling proof from $H = 0$ to $H = B^{p/2}$. The flow makes H, T continuous, while the uniform estimate $T/B \rightarrow \Theta_* \neq 0$ keeps the denominator nonzero. Thus the continuous normalized coordinate satisfies

$$\frac{H}{|T|^p} = 0 \quad \text{at the first endpoint}, \quad \frac{H}{|T|^p} \asymp B^{-p/2} \longrightarrow +\infty \quad \text{at the second}.$$

It therefore traverses the interior of $[1.2679251755, 1.2679451755]$. The preimage of that interior is a nonempty open set and hence contains an open interval. Restricted to this preimage, compactness of the projective-amplitude interval, uniform endpoint localization, and differentiated regularized convergence supply the remaining finite-mass cusp hypotheses. The preceding theorem classifies every member of this open set as collision or escape. $\square$

This boundary emanates from the massless planar joint-limit collision and persists in the surrounding finite-mass section family. It is not a tied finite-mass Pythagorean orbit or a brake, and no positive-mass classical collision is continued physically. The preceding theorem classifies a neighborhood of this first root. What remains is exclusion or certification of the remaining collision zeros and outgoing control on the compact components. In particular, the exact tied trajectory selects only one point $(\hat{\kappa}_B, \delta_B)$ on the fixed section; the collision-boundary persistence corollary does not

prove that it solves $\widehat{\kappa}_B = \widehat{\kappa}_c(B, \delta_B)$. The theorem replaces this section-dependent condition by the intrinsic scalar gap

$$\Delta_{\text{intr}}(B) = \frac{H_B}{|T_B|^{\mu/\tau}} - \kappa_c(B, T_B, S_B, \Xi_B).$$

No componentwise sign classification of this gap is proved here. The real collision-sampling theorem gives only a cyclewise dichotomy between an earlier collision and a zero of this selected graph. Its isolated roots are not proved rational.

Parameterize the restricted incoming parabolic curve by its asymptotic phase intercept χ , normalized by $\chi = \lim_{Y \rightarrow \infty} (\phi_Y + (4/3)Y^{3/2})$ modulo 2π . This coordinate includes the finite restricted quadrupole correction between infinity and a fixed section; a raw monopole section phase does not. The incoming-tail theorem inserts the true full regularized state, producing infinitely many open terminal nonperiodicity intervals accumulating at $B = 0$ result. Strict restricted escape on a compact intercept arc transfers to a finite outgoing scaled section; there $e = -C/B + o(B^{-1})$ and $\dot{\rho} = sB^{-1/2} + o(B^{-1/2})$, so the terminal escape lemma applies uniformly, yielding escape or an earlier classical inner collision and excluding a later brake in either case.

Theorem 37 (Infinite subfamily). *There are infinitely many open real Euclid-parameter intervals accumulating at $u = 0$ on which the Pythagorean–Burrau solution is nonperiodic. In particular, infinitely many distinct primitive Pythagorean triples satisfy the nonperiodicity conjecture.*

Proof. Shrink a restricted escape-intercept arc by a fixed positive margin. The incoming-tail theorem and robust terminal transfer cover every sufficiently small preimage interval. A trajectory that collides before section reach is already nonperiodic; every other trajectory satisfies the terminal escape-or-collision certificate. The exact derivative $\Phi'_{\text{ref}} \sim -(3\pi/2)B^{-5/2}$ gives infinitely many disjoint preimage intervals. Since $B(u) = 2u/(1 + u^2)$ is a homeomorphism near zero and rational u is dense, each contains a rational Euclid parameter. Reduction by the simultaneous mass–length scaling gives distinct primitive triples. $\square$

The existential rational consequence can be sharpened to one explicit primitive family.

Theorem 38 (Positive density in an explicit family). *Let*

$$(a_n, b_n, c_n) = (4n^2 - 1, 4n, 4n^2 + 1), \quad n \geq 1.$$

There is a positive-lower-natural-density set of indices n for which the corresponding Pythagorean–Burrau solution is nonperiodic.

Proof. Here $u_n = 1/(2n)$ and $B_n = 4n/(4n^2 + 1)$. Direct substitution in the exact reference clock gives

$$F(x) := \frac{\Phi_{\text{ref}}(B_x)}{2\pi} = \frac{(2x - 1)^{3/2}(2x + 1)(4x^2 + 1)^{3/4}}{32x^{5/2}},$$

and therefore

$$F''(x) = \frac{3}{8}x^{-1/2} + \frac{1}{32}x^{-3/2} + O(x^{-5/2}).$$

On each dyadic block, van der Corput’s second-derivative estimate makes every nontrivial Weyl sum $O_h(N^{3/4})$. Weyl’s criterion proves that $F(n)$ is equidistributed modulo one. Consequently every fixed compact escape-intercept arc J selects a set of indices of density $|J|/(2\pi)$. The incoming-tail and terminal-transfer theorems prove every sufficiently large selected member nonperiodic. Primitivity follows from $\gcd(4n, 4n^2 - 1) = 1$. $\square$

9 Cardinality and the remaining obstruction

The rational parameters are countable and dense. Neither countability, measure zero, analyticity, nor the nominal three-equations/two-unknowns count excludes rational zeros. A proof still requires either no real zero, an exact arithmetic characterization of all real zeros, or a finite exact reduction.

The active compact-middle route follows the strict-maximum branches $\dot{I} = 0$, on which the exact shape-speed factorization reduces a possible brake to $\zeta = 0$, equivalently $K = U - U_0 = 0$. It seeks to classify these branches and prove their shape speed never vanishes. The earlier local route factors a possible brake into two radial-rate equations followed by an angular-slip equation; it remains useful where its two-by-two event Jacobian is nonsingular. Ordinary scans show both signs of angular slip on different double-radial events, ruling out a universal fixed-sign shortcut. The skinny route now controls the full incoming plunge and proves open escape windows. On the finite-turn side, fixed-resonance first-order matching and a uniform outgoing angular-momentum theorem exclude a brake at the first outer turn throughout the infinite accumulating resonance layer. The returning-tail estimate propagates the sign back to a fixed large section; the next problem is to classify the resulting compact collision-regularized central-encounter map.

10 Necessary integral identities

For a hypothetical second brake at τ , the Lagrange–Jacobi identity is

$$\ddot{I} = 4K - 2U = 2U - 4U_0.$$

Both endpoint dilational momenta vanish, so

$$\int_0^\tau U dt = 2U_0\tau, \quad \int_0^\tau K dt = U_0\tau.$$

Integrating each labelled acceleration also gives zero. If $d_{ij} = q_j - q_i$, multiplication by the fixed initial positions yields the additional scalar condition

$$\int_0^\tau \sum_{i < j} m_i m_j \frac{d_{ij}(0) \cdot d_{ij}(t)}{r_{ij}(t)^3} dt = 0.$$

Its integrand is initially positive, but no global projection inequality has been proved through close encounters.

There is a complementary exact torque condition. Orient $r_{12} = q_2 - q_1$, $r_{23} = q_3 - q_2$, $r_{31} = q_1 - q_3$, let $\Delta_2 = (q_2 - q_1) \times (q_3 - q_1)$, and set $\ell_{ij} = r_{ij} \times \dot{r}_{ij}$. Then

$$\begin{aligned} \dot{\ell}_{12} &= m_3 \Delta_2 (r_{23}^{-3} - r_{31}^{-3}), \\ \dot{\ell}_{23} &= m_1 \Delta_2 (r_{31}^{-3} - r_{12}^{-3}), \\ \dot{\ell}_{31} &= m_2 \Delta_2 (r_{12}^{-3} - r_{23}^{-3}). \end{aligned}$$

At the Pythagorean launch in the strict fundamental interval their signs are $(-, +, -)$. A second brake forces the integral of every right-hand side over $[0, \tau]$ to vanish. This reduces a torque obstruction to controlling syzygy orientation and distance-order changes; neither cancellation has yet been excluded globally.

More precisely, continuity and the strict initial signs imply that every one of the three torque integrands must attain the opposite strict sign before a second brake. As long as $\Delta_2 > 0$, this

requires, respectively, $r_{23} < r_{31}$, $r_{31} > r_{12}$, and $r_{12} < r_{23}$ at possibly different times. Since initially $r_{12} > r_{23} > r_{31}$, a second brake before the first syzygy requires all three pairwise side-order reversals. This necessary condition does not prevent those crossings or the common sign reversal at a syzygy.

These angular momenta also give a second reduced brake residual. For total mass M , weighted position Q , momentum P , and angular momentum L ,

$$m_1 m_2 \ell_{12} + m_2 m_3 \ell_{23} + m_3 m_1 \ell_{31} = ML - Q \times P.$$

At $P = L = 0$, vanishing of ℓ_{12}, ℓ_{23} therefore kills the third. For a noncollinear configuration, three radial pair velocities compatible with triangle closure have one common homothetic rate λ ; then $\dot{I} = 2\lambda I$. Consequently

$$(\ell_{12}, \ell_{23}, \dot{I}) = 0 \iff \text{a labelled brake}$$

away from syzygy. On a regular nonsyzygy I -maximum only the two signed angular momenta remain. The equivalence is false at syzygy, where the Hopf residual must be retained.

This is an exact recharting of the previous shape residual. Rotate so that $X = (R, 0)$ and $Y = (C, D)$, and set $\alpha = m_1/(m_1 + m_2)$, $\kappa = \mu_1/\mu_2$. Solving $\dot{I} = L = 0$ for V in terms of $U = (u_x, u_y)$ gives $\zeta = 2\mu_1 R(u_x + iu_y)$ and

$$\det \frac{\partial(\ell_{12}, \ell_{23})}{\partial(u_x, u_y)} = -\alpha R D \left(1 + \frac{\kappa R^2}{C^2 + D^2} \right).$$

Every factor except $RD = X \times Y$ has fixed positive sign. Hence the torque and ζ planes have the same zero set and the torque chart degenerates precisely at syzygy.

The two side-order gaps initially point into a simple candidate cone. Direct calculation gives

$$(r_{12} - r_{23})''(0) = -\frac{S_9(u)}{2u(1-u)^2(1+u)(1+u^2)^2},$$

where $S_9 = u^9 - 9u^8 + 10u^7 - 10u^6 - 8u^5 - 16u^4 - 2u^3 + 2u^2 - u + 1$. Its ten degree-9 Bernstein coefficients on $[0, 5/12]$ are all positive. Moreover

$$(r_{23} - r_{31})''(0) = -\frac{(u^2 + 2u - 1)(3u^2 - 2u + 1)[u^2(u - 1)^2 + (u + 1)^2]}{2u^2(1 - u)^2(1 + u^2)^2}.$$

Hence the first acceleration is negative and the second positive on the strict fundamental interval. The suggested persistence $r_{12} > r_{23} > r_{31}$ to first syzygy remains conjectural.

The gravitational part of the second gap does have a global sign. Write

$$\ddot{r}_{ij} = \frac{\ell_{ij}^2}{r_{ij}^3} + \mathfrak{g}_{ij}.$$

After scaling $r_{12} = 1$, parameterize every strict ordered triangle by

$$r_{23} = 1 - \frac{tw}{2}, \quad r_{31} = 1 - t + \frac{tw}{2}, \quad 0 < t, w < 1,$$

and put $u = (\sqrt{2} - 1)v$, $0 < v < 1$. The exact rationalized expression for $\mathfrak{g}_{23} - \mathfrak{g}_{31}$ has a positive denominator and a tensor-degree $(7, 6, 2)$ numerator. Its 168 Bernstein coefficients over the unit cube are all nonnegative, with 137 strictly positive. Thus

$$\mathfrak{g}_{23} - \mathfrak{g}_{31} > 0$$

throughout the open ordered cone. The still-open term is the centrifugal difference $\ell_{23}^2/r_{23}^3 - \ell_{31}^2/r_{31}^3$, which can be negative on the actual orbit.

The first gravitational gap has the complementary sign on the ordered right/obtuse subcone. In the displayed coordinates,

$$r_{23}^2 + r_{31}^2 - 1 = \frac{t^2 w^2 - 2t^2 w + 2t^2 - 4t + 2}{2}.$$

Its minimum over $w \in [0, 1]$ shows that right/obtuse shapes require $t > 2 - \sqrt{2}$, except at the symmetric endpoint. Substitute $t = 2 - \sqrt{2} + (\sqrt{2} - 1)s$. The exact tensor-degree $(7, 6, 2)$ numerator of $\mathbf{g}_{12} - \mathbf{g}_{23}$ has 168 nonpositive Bernstein coefficients, 150 of them negative, and a positive denominator. Therefore

$$\mathbf{g}_{12} - \mathbf{g}_{23} < 0$$

throughout the strict ordered right/obtuse cone. Gravity thus points into both side-order faces; persistence of obtuseness and domination of the two centrifugal differences remain open.

The torque history itself reduces to a scalar weighted average. On a positive-area ordered segment put $R = r_{12}$, $p = r_{23}$, $q = r_{31}$ and

$$F = \Delta_2(q^{-3} - R^{-3}), \quad G = \Delta_2(p^{-3} - R^{-3}), \quad k = \frac{G}{F}.$$

If $\mathcal{A} = \int_0^t F ds$ and $\mathcal{C} = \int_0^t G ds$, then

$$\eta = \frac{\mathcal{C}}{\mathcal{A}} = \frac{m_1 - \ell_{31}}{m_2 \ell_{23}}, \quad \dot{\eta} = \frac{F}{\mathcal{A}}(k - \eta).$$

Thus strict decrease of k would imply $k(t) < \eta(t) < k(0)$. Writing $x = p/R$, $y = q/R$ gives

$$k = \frac{y^3(1 - x^3)}{x^3(1 - y^3)}.$$

There is an exact first-order threshold hidden in these variables. In shape time $ds = dt/R^{3/2}$, put $W = \ell_{23}/\sqrt{R}$ and $\delta = \Delta_2/R^2$. Exact elimination of normalized shape velocity gives

$$(\log k)_s = \frac{W}{\delta} \mathcal{A}(u, x, y)[\eta - h(u, x, y)].$$

Here $h = y^2 N/(x^2 B)$, where

$$\begin{aligned} B &= -m_2 x^5 - m_2 x^3 y^2 + m_2 x^3 + m_2 x^2 + 2m_2 y^5 - m_2 y^2 - m_2 \\ &\quad + x^5 y^2 - x^3 y^4 + x^3 y^2 - x^2 y^5 + y^7 + y^5 - 2y^2, \\ N &= 2m_1 x^5 - m_1 x^2 y^3 - m_1 x^2 - m_1 y^5 + m_1 y^3 + m_1 y^2 - m_1 \\ &\quad + x^7 - x^5 y^2 + x^5 - x^4 y^3 + x^2 y^5 + x^2 y^3 - 2x^2, \end{aligned}$$

and

$$\mathcal{A} = \frac{-3B}{2m_1 y^2 (x-1)(y-1)(x^2+x+1)(y^2+y+1)}.$$

On the complete ordered shape-mass cube, $-B$ has a degree- $(7, 6, 2)$ numerator with 168 exact Bernstein coefficients, 134 strict and 34 zero, with the denominator sign accounted for in the exact rational substitution. Direct sign reconstruction gives $\mathcal{A} > 0$ and

$$(\log k)_s < 0 \quad \Longleftrightarrow \quad \eta < h.$$

At the tied launch, $h - k(0) > 0$ by exact factorization. Thus the monotonicity problem is precisely a scalar history noncontact problem for the graph $\eta = h$, rather than a sign problem for an uncontrolled velocity Hessian. In fact $h > k$ throughout the complete strict ordered cone. After the same cube substitution, its degree-(11, 11, 2) core has 432 exact Bernstein coefficients, 317 strict and 115 zero, with the denominator sign handled exactly.

At a contact $\eta = h$, let $X = \delta x_s / W$, $Y = \delta y_s / W$ and set

$$S = h_x X + h_y Y, \quad P = m_1 \delta^2 (y^{-3} - 1)(k - h) < 0.$$

The contact derivative reduces exactly to

$$W \delta(\eta - h)_s = P - ZS.$$

Thus every contact with $S \geq 0$ is automatically safe. On $S < 0$ the sharp condition is the single amplitude inequality

$$Z < Z_J = \frac{P}{S}.$$

The energy-normalized $u = 2/5$ ambient contact below has $Z/Z_J = 1.05745644351720\dots > 1$. Hence total energy misses the required bound by a definite amount, while the brake-reachable amplitude inequality on $\{\eta = h, S < 0\}$ is a materially smaller target than global curvature domination. The torque-contact threshold Z_J and the second-gap threshold Z_2 do not globally dominate one another: exact rational examples give both $Z_J/Z_2 < 1$ and $Z_J/Z_2 > 1$. A common amplitude theorem must therefore use the appropriate minimum on their dangerous regions.

The history comparison also admits an exact forced-lag form. Define

$$d = h - k > 0, \quad e = \eta - k, \quad w = h - \eta,$$

and

$$\lambda = \frac{m_1 \delta (y^{-3} - 1)}{W} > 0, \quad c = \frac{kW\mathcal{A}}{\delta} > 0.$$

Then

$$e_s + (\lambda + c)e = cd, \quad w_s + (\lambda + c)w = d_s + \lambda d.$$

In particular,

$$e(s) = \int_0^s c(\tau) d(\tau) \exp\left(-\int_\tau^s (\lambda + c) d\xi\right) d\tau.$$

The endpoint issue in this reduction is essential. Put $a = \lambda + c$ and, for a fixed interior reference point $s_0 > 0$, define

$$M(s) = \exp\left(\int_{s_0}^s a(\xi) d\xi\right).$$

Then $(Mw)_s = Mq$, where $q = d_s + \lambda d$. The reference point cannot be launch: W has a simple zero there and $\lambda \sim 1/s$. Moreover, at a transverse ordered terminal syzygy $s = S$, $\delta \rightarrow 0$ and $c = kW\mathcal{A}/\delta$ can diverge. Thus $w(S) = 0$ alone does not imply $M(s)w(s) \rightarrow 0$.

A correct sufficient statement is the following: if w is positive near launch, q is positive before one switch and negative thereafter, and

$$\liminf_{s \uparrow S} M(s)w(s) \geq 0,$$

then $w > 0$ throughout the interior. Indeed Mw first increases and then strictly decreases to a nonnegative limit. This additional terminal bound has not been established globally for the tied family.

The omitted bound cannot be inferred from the scalar equation. On $0 < s < 1$, choose

$$a = \frac{1}{s} + \frac{1}{1-s}, \quad w = (1-s)(1+s-3s^2), \quad q = (1-s) \left(\frac{1}{s} + 2 - 9s \right).$$

Here $w_s + aw = q$, $w(0) = 1$, and $w(1) = 0$. The forcing has exactly one positive-to-negative switch, but $w < 0$ near 1. Up to a positive constant, $M = s/(1-s)$ and $Mw \rightarrow -1$. This is a counterexample to the unqualified scalar inference, not a Newtonian orbit or a counterexample to the Pythagorean conjecture. The Volterra formula for e above is understood as an improper integral with lower limit tending to zero; its kernel only integrates from $\tau > 0$ to $s < S$.

The normalized forcing has the rational expression

$$E := W\delta q = ZD + m_1\delta^2(y^{-3} - 1)d, \quad D = d_x X + d_y Y.$$

At $\eta = h$, $E = ZS - P$, so this formulation contains the sharp contact amplitude threshold. Neither $E \geq 0$ nor pointwise one-way crossing of $E = 0$ holds on the corresponding ambient energy domain. An exact rational state at $u = 2/5$, $(x, y) = (1957/2000, 323/2000)$, $\eta = 3/80$, and normalized contraction $\gamma = -1/100$ has ordered obtuse shape, $k < \eta < h$, decreasing area and inertia, exact Burrau energy after a positive scale change, and $E = 0$, but $W\delta E_s > 0$. It is not known reachable from the brake. A likewise exact energy-compatible witness at $u = 207/500$ disproves invariance of the buffered barrier $h - \eta + y(1 - Z/Z_J)$. These witnesses localize the missing hypothesis: the proof must use integrated brake history rather than the listed static phase-space inequalities.

The terminal ordered-syzygy face has a stronger exact structure that is not present in the interior.

Theorem 39 (Ordered-syzygy amplitude ordering). *Let $(m_1, m_2, m_3) = (m, n, 1)$ be tied masses with $0 < u \leq \sqrt{2} - 1$, normalize $r_{12} = 1$, and let body 3 lie between bodies 1 and 2. Put $q = r_{31}$, $p = r_{23} = 1 - q$, and assume the strict torque-compatible range $0 < q < n/(m+n)$. The torque graph and the collinear momentum constraints satisfy*

$$h(m, n, p, q) = \eta_{\text{col}} = \frac{q(m+p)}{p(n+q)}.$$

The quotient $Z_J = P/S$ has a path-independent rational extension Z_J^{syzy} to this face, and

$$\boxed{1 < Z_J^{\text{syzy}} < Z_2}.$$

Proof. Direct symbolic division shows that P and S contain the same exact twice-area-squared factor. After its cancellation, substitution of $p = 1 - q$ gives

$$Z_J^{\text{syzy}} = \frac{2m^2(n+q)^2(q-1)(q^2-q+2)N_J}{q\{mn + mq^2 + n(1-q)^2\}D_J},$$

where N_J, D_J are explicit polynomials of q of degree five. Moreover

$$C_2 = \frac{-E_2}{m^2q(n+q)^2(q-1)^3}, \quad Z_2 - Z_J^{\text{syzy}} = \frac{-m^2(n+q)^2(q-1)(2q-1)Q}{q\{mn + mq^2 + n(1-q)^2\}E_2D_J}.$$

Set

$$u = (\sqrt{2} - 1)v, \quad q = \frac{nz}{m+n}, \quad 0 < v \leq 1, \quad 0 < z < 1.$$

After clearing positive mass and mass-sum denominators, the cores N_J, D_J, E_2, Q and the numerator of $Z_J^{\text{syzy}} - 1$ have bidegrees

$$(11, 5), \quad (12, 5), \quad (11, 3), \quad (26, 10), \quad (27, 10).$$

Their exact $\mathbb{Q}(\sqrt{2})$ tensor-Bernstein coefficients consist, respectively, of

$$71 - /10, \quad 71 + /70, \quad 39 - /90, \quad 274 + /230, \quad 285 + /230.$$

Thus $N_J < 0$, $D_J > 0$, $E_2 < 0$, and $Q > 0$ in the open square. On $v = 1$ the edge coefficients have the same strict sign except only at $z = 1$, so the conclusion includes the equal-mass edge. Since $q < 1/2$, the remaining factors have manifest signs. Hence $Z_J^{\text{syzy}} > 1$, $C_2 < 0$, and $Z_2 - Z_J^{\text{syzy}} > 0$; the gravitational factor in Z_2 is positive by the ordered-cone gravity theorem. $\square$

Consequently the correct torque-history target is strict separation $\eta < h$ in the positive-area interior with forced equality at terminal syzygy. Although the interior needs the appropriate minimum of two thresholds, only $Z < Z_J^{\text{syzy}}$ remains as the terminal amplitude obligation. This theorem orders the barriers but does not prove that brake-generated histories obey that inequality.

The threshold geometry at the equal-mass corner can be resolved exactly. For fixed tied masses, $\partial_q R_J < 0$ on the torque-compatible syzygy face. Its derivative denominator is $q^2(1 - q)^2(m^2 n^2 + 1)D_J^2 > 0$; the cleared bidegree-(30, 12) numerator has 374 negative and 29 zero exact Bernstein coefficients. At the singular corner put $v = 1 - \epsilon$, $z = 1 - \lambda\epsilon$, and $\kappa = (28 - 3\sqrt{2})/11$. Then

$$\lim Z_J^{\text{syzy}} = \frac{Z_0 + \kappa\lambda Z_\infty}{1 + \kappa\lambda}, \quad \lim R_J = \frac{R_0 + \kappa\lambda R_\infty}{1 + \kappa\lambda},$$

where

$$Z_0 = \frac{21(1 + \sqrt{2})}{22}, \quad Z_\infty = \frac{3836 + 2709\sqrt{2}}{6128},$$

$$R_0 = \frac{11 + 2\sqrt{2}}{55}, \quad R_\infty = \frac{-361 + 2273\sqrt{2}}{3830}.$$

Thus $Z_0 > Z_\infty > 1$ and $R_0 < R_\infty$; the two iterated corner limits do not commute.

Theorem 40 (Near-isosceles terminal amplitude). *There exists $v_0 < 1$ such that the continued first syzygy of every tied member with $v_0 < v < 1$ lies on the torque-compatible ordered face and satisfies*

$$Z < 1 < Z_J^{\text{syzy}}.$$

Proof. In the symmetric endpoint reduction, a pinned interval Taylor cover proves that the first crossing $h = 0$ is unique, transverse and collision-free, and that $Z_* = xh^2/2 < 1$. A separate C^1 Poincare-map enclosure propagates the exact tied launch tangent $(m_{1,v}, m_{2,v}, q_{3x,v}) = (-1/2, 1/2, 1/\sqrt{2})$ and proves, for $z(v) = q(v)(m(v) + n(v))/n(v)$,

$$z_v(1) > 18 > 0.$$

The implicit-function theorem therefore gives $z(v) = 1 - z_v(1)(1 - v) + O((1 - v)^2) < 1$ and $Z(v) < 1$ for $v < 1$ sufficiently close to one. The uniform exact bound $Z_J^{\text{syzy}} > 1$ completes the claim. $\square$

Theorem 41 (Near-isosceles pre-syzygy noncontact). *There exists $v_0 < 1$ such that, for every $v_0 < v < 1$, the tied solution is collision-free through its continued first syzygy $\tau(v)$ and satisfies*

$$r_{12} > r_{23} > r_{31}, \quad \eta < h, \quad (\log k)_s < 0 \quad (0 < t < \tau(v)).$$

At $t = \tau(v)$ one has $\eta = h$, the crossing is transverse, and no labelled brake occurs on $(0, \tau(v)]$.

Proof. At the symmetric endpoint put $q_1 = (-x, 0)$, $q_2 = (x, 0)$, $q_3 = (0, H)$ and $r = (x^2 + H^2)^{1/2}$. Reflection and leg-swap symmetry restrict the odd tied tangent to

$$q_{1,v} = (\alpha, b), \quad q_{2,v} = (\alpha, -b), \quad q_{3,v} = (\gamma, 0), \quad c = \gamma - \alpha.$$

Direct variation of Newton's equations gives

$$\begin{aligned} \ddot{c} &= \frac{x}{r^3} - \frac{1}{8x^2} + (1 + \sqrt{2}) \frac{(2x^2 - H^2)c - 3Hxb}{r^5}, \\ \ddot{b} &= -\frac{3Hxc}{r^5} + \left(\frac{2H^2 - x^2}{r^5} - \frac{\sqrt{2}}{8x^3} \right) b, \end{aligned}$$

with launch data $(x, H, c, b) = (1/2, 1/2, 1/\sqrt{2}, 0)$ and zero tangent velocities.

Put

$$a = H\dot{x} - x\dot{H}, \quad a_v = c\dot{H} + b\dot{x} - x\dot{b} - H\dot{c}, \quad d = \frac{r}{2x}, \quad Y = \frac{xc - Hb}{2xr}.$$

The normalized side derivatives are $(p_v, q_v) = (-Y, Y)$. Exact differentiation of the rational threshold, including $(m_v, n_v) = (-1/2, 1/2)$, gives

$$g(t) := \partial_v(\eta - h)(1, t) = -\sqrt{2} - 2\frac{a_v}{a} - A(d)Y - B(d),$$

where

$$A(d) = -\frac{4(2\sqrt{2}d^2 + 1)(16d^5 - 5d^3 - 4d^2 + 2)}{d(d-1)(4d^2 + \sqrt{2})^2(d^2 + d + 1)}, \quad B(d) = -\frac{2}{4d^2 + \sqrt{2}}.$$

The quotient extends analytically through launch and exact initial rates give

$$g(0) = \frac{32 + 8\sqrt{2}}{7} > 0.$$

A pinned interval Taylor cover of the reduced orbit and tangent proves $g > 6$ on $0 < t \leq 10^{-3}$ using integral averages of $a_t, a_{v,t}$, $g > 4$ on $10^{-3} \leq t \leq 0.43$, and $g_t < -6$ from 0.43 through the unique syzygy step. Every full-step tube also gives $d < 1$ and $Y > 0$; on the compact part it gives $d < 3/4$ and $Y > 1/2$. The reduced equations and the differentiated threshold were independently regressed against the full Cartesian force Jacobian and the generic symbolic threshold.

Let $F(v, t) = \eta(v, t) - h(v, t)$ after analytic launch extension. Reflection gives $F(1, t) = 0$. The exact ordered-syzygy identity gives $F(v, \tau(v)) = 0$ on the continued torque-compatible event, hence $g(\tau(1)) = 0$. The terminal derivative enclosure and the preceding positive cover prove

$$g(t) > 0 \quad (0 \leq t < \tau(1)), \quad g(\tau(1)) = 0, \quad g_t(\tau(1)) < 0.$$

Analytic division first gives $F = (v - 1)F_1$ with $F_1(1, t) = g(t)$. Near the event, division by its transverse graph gives

$$F(v, t) = (v - 1)(t - \tau(v))F_2(v, t), \quad F_2(1, \tau(1)) = g_t(\tau(1)) < 0.$$

Compactness away from the event and this terminal factorization imply $F < 0$ for all $v < 1$ sufficiently close to one and all $0 < t < \tau(v)$. Similarly $d < 1$ and $Y > 0$ imply the displayed strict side ordering. Positive area and that ordering keep a pair angular momentum strictly nonzero, excluding a brake; the exact threshold identity supplies the claimed sign of $(\log k)_s$. $\square$

The two near-isosceles theorems close the interior torque contact and the terminal first-syzygy amplitude condition on the complete first arc in a one-sided endpoint neighborhood. They do not exclude a later brake after the unfolded close base-pair encounter.

That encounter itself admits a regular transverse classification. Let $g = q_2 - q_1$, let $G = q_3 - C_{12}$ be the complementary Jacobi coordinate, and set

$$g = w^2, \quad \frac{dt}{d\sigma} = |w|^2, \quad z = \frac{dw}{d\sigma}.$$

For $d_{31} = G + ng/(m+n)$ and $d_{23} = G - mg/(m+n)$, direct force reduction gives

$$\ddot{g} = -\frac{(m+n)g}{|g|^3} + \frac{d_{23}}{|d_{23}|^3} - \frac{d_{31}}{|d_{31}|^3},$$

$$\ddot{G} = -\frac{m+n+1}{m+n} \left(\frac{md_{31}}{|d_{31}|^3} + \frac{nd_{23}}{|d_{23}|^3} \right).$$

Writing $e = |\dot{g}|^2/2 - (m+n)/|g|$ and $F = d_{23}/|d_{23}|^3 - d_{31}/|d_{31}|^3$, the forced LC equations include

$$w' = z, \quad z' = \frac{e}{2}w + \frac{|w|^2}{2}\bar{w}F, \quad G' = |w|^2P, \quad P' = |w|^2\ddot{G}, \quad t' = |w|^2.$$

Theorem 42 (Near-isosceles binary unfolding). *There exists $v_1 < 1$ such that every tied member with $v_1 < v < 1$ traverses the encounter corresponding to the endpoint's base-pair collision as a classical collision-free near miss. For a nonzero constant χ ,*

$$\min_{\text{enc}} r_{12}(v) = \chi^2(1-v)^2 + o((1-v)^2), \quad \chi^2 > 900.$$

At the LC collision section its pair angular momentum has reversed its negative launch sign: $\ell_{12} > 0$.

Proof. At $v = 1$ the LC launch state is

$$w = 1, \quad z = 0, \quad e = -\sqrt{2}, \quad G = i/2, \quad P = 0, \quad m = n = 1/\sqrt{2}.$$

The collision is $w_r = 0$. A pinned interval C^1 Poincare calculation proves at this section

$$z_r < -\frac{4}{5}, \quad r_{31}^2, r_{23}^2 > \frac{1}{500}, \quad \chi := \partial_v w_i < -30.$$

The tied launch tangent used in this calculation is exact: $G_{x,v} = 1/(2\sqrt{2})$ and $(m_v, n_v) = (-1/2, 1/2)$, with every other LC launch derivative zero. Cartesian force regression independently verifies the Jacobi and LC fields.

On the symmetric orbit $w_i = z_i = 0$. Hence the Jacobian of $(\sigma, v) \mapsto (w_r, w_i)$ at the collision has determinant $z_r \chi \neq 0$. The inverse-function theorem isolates the collision zero, while the positive unselected separations and compact continuity exclude any other collision in the encounter block. On the graph $w_r = 0$, $w_i = \chi(v-1) + O((v-1)^2)$. Since $z_i = 0$ on the base orbit, the minimizer of $|w|^2 = r_{12}$ shifts by only $O((v-1)^2)$ in regularized time, giving the stated asymptotic. Finally

$$\ell_{12} = 2(w_r z_i - w_i z_r), \quad \partial_v \ell_{12}(1) = -2\chi z_r < 0.$$

Thus $v-1 < 0$ implies $\ell_{12} > 0$ at the section. $\square$

This theorem concerns the nearby classical trajectories. The endpoint $v = 1$ still terminates at its binary collision; its LC continuation is used only as an analytic comparison orbit.

Theorem 43 (Punctured near-isosceles nonperiodicity). *There exists $v_* < 1$ such that every real tied member with $v_* < v < 1$ is nonperiodic. It is collision-free and has no labelled brake through a fixed LC section; thereafter it either ends in a classical binary collision or has body 3 escape. Consequently the same conclusion holds for infinitely many primitive Pythagorean triples.*

Proof. The exact LC energy constraint is

$$2|z|^2 - (m + n) - e|w|^2 = 0.$$

On the symmetric endpoint orbit it gives $2|z|^2 = \sqrt{2}$ whenever $w = 0$, so every selected collision is transverse. Pinned oriented C^1 Poincare maps enclose four such zeros before $\sigma = 7$ and put the fifth strictly after 7. At each of the first four zeros, the tied normal derivative $(w_i)_v$ is bounded away from zero. Thus every Jacobian

$$\frac{\partial(w_r, w_i)}{\partial(\sigma, v)}$$

is nonsingular. The inverse-function theorem isolates all four zeros, while a 7000-step C^0 cover proves

$$r_{31}^2, r_{23}^2 > \frac{1}{500} \quad (0 \leq \sigma \leq 7).$$

Compactness outside the four collision boxes now gives a punctured neighborhood in which every nearby tied orbit is collision-free through $\sigma = 7$.

At an ordinary point of this LC chart,

$$\dot{g} = \frac{2z}{\bar{w}}, \quad \dot{G} = P.$$

Hence, after center-of-mass reduction, a labelled brake is equivalent to $z = P = 0$. The polynomial residual $\mathcal{R}_{LC} = |z|^2 + |P|^2$ remains regular at comparison collisions. At the $\sigma = 1/2$ interface the cover gives $G_y > 1/10$, so the unique first syzygy lies later. Every complete interval tube from that overlapping pre-syzygy interface to $\sigma = 7$ satisfies $\mathcal{R}_{LC} > 2$. Compact continuity transfers this inequality to the nearby classical orbits; the preceding pre-syzygy theorem covers the initial segment.

At $\sigma = 7$, take $\eta = 4$ in the terminal binary–escaper theorem. The same interval cover proves, with $M = m + n = \sqrt{2}$, $R = M/4$ and $s = |G| - R$,

$$s > 1, \quad \dot{\rho} > 2, \quad \delta = \frac{\dot{\rho}^2}{2} - \frac{M+1}{s} > \frac{1}{20},$$

and

$$-4 - e - \frac{\sqrt{2MR}}{\sqrt{2\delta} s^2} > 2.$$

All inequalities are strict and therefore persist in a tied neighborhood. The terminal theorem says that each future trajectory either suffers an inner classical collision or escapes with uniformly positive outward radial speed. Neither possibility permits a later brake. Rational Euclid parameters are dense in the resulting open u interval, giving infinitely many primitive triples. $\square$

The cutoff v_* is existential; making it effective would quantify a Pell subfamily but is not required for the interval theorem. The full conjecture still requires control of the middle parameter range.

The remaining terminal inequality admits a useful exact coercive form. Restore $R = r_{12}$ and set

$$D = mn + mq^2 + n(1 - q)^2, \quad E = (m + n)q - n, \quad M = m + n + 1.$$

With $\sigma = R_s/R$ and $Q = q_s$, longitudinal momentum reduction gives

$$2K_{\parallel}R = \frac{D}{M} \left(\sigma + \frac{EQ}{D} \right)^2 + \frac{mnQ^2}{D} = \frac{R\dot{I}^2}{4I} + \frac{mnQ^2}{D}, \quad I = \frac{R^2D}{M}.$$

The transverse kinetic energy is $2K_{\perp}R = FZ$, where

$$F = \frac{nD}{m(1 - q)^2(n + q)^2}.$$

Consequently energy is the exact identity

$$FZ + 2K_{\parallel}R = 2(U - U_0R), \quad U = mn + \frac{n}{1 - q} + \frac{m}{q}.$$

If

$$R_J = \frac{U - FZ_J^{\text{syzy}}/2}{U_0},$$

then

$$Z < Z_J^{\text{syzy}} \iff 2K_{\parallel}R > 2U_0(R_J - R).$$

Natural terminal derivative signs cannot furnish the required positive lower bound. Indeed put $H = n - (m + n)q > 0$ and choose

$$\frac{1 - 2q}{2} < c < \frac{D}{H};$$

this interval is nonempty because

$$2D - H(1 - 2q) = 2mn + (m - n)q + n > 0.$$

For $\sigma = -\varepsilon$, $Q = -c\varepsilon$, one has strict decrease of r_{31} , strict increase of $r_{23} - r_{31}$, and $\dot{I} < 0$, whereas $2K_{\parallel}R = O(\varepsilon^2)$. Thus even the full set of qualitative terminal signs permits ambient energy states with $Z > Z_J^{\text{syzy}}$ when $R < R_J$. Quantitative information inherited from the brake history, rather than terminal monotonicity alone, is indispensable.

For $J = \log k$, the purely gravitational contribution to $\ddot{J}$ is strictly negative throughout the ordered right/obtuse cone. After the same (s, w, v) substitution, its degree-(9, 9, 2) numerator has 300 nonnegative exact Bernstein coefficients, 268 positive, while its denominator is

$$-4096(1 + u^2)p^3q^3(1 - p^3)(1 - q^3) < 0.$$

At the brake the velocity term vanishes and

$$\ddot{J}(0) = \frac{3(1 + u^2)^4(u^2 + 2u - 1)(2u^4 - 3u^3 + 3u^2 + u + 1)}{u^3(1 - u)^3(u^4 + 3)(u^4 + 2u^3 + 6u^2 + 2u + 1)} < 0.$$

The total curvature contains an additional velocity-Hessian term. Its sign is not controlled by current-state inequalities. The first-order threshold above supersedes direct curvature domination as the primary formulation: the missing lemma is the brake-history inequality $\eta < h$.

The qualifier “reachable-state” is essential even at a hypothetical first return of J' to zero. For the exact tied parameter $u = 2/5$ and strict ordered obtuse shape

$$(R, p, q) = \left(1, \frac{39}{40}, \frac{1}{30}\right),$$

exactly solving $L = J' = 0$ gives a velocity with

$$(\ell_{12}, \ell_{23}, \ell_{31}) = \left(-1, \frac{28914232361390649}{36366761131529200}, -\frac{122841961770469}{1818338056576460}\right).$$

Thus all three pre-syzygy pair-angular-momentum signs hold, but the velocity curvature is positive. At scale $R = 1/2$, after choosing its amplitude from $H = -U_0$, exact arithmetic gives

$$R^3 J'' = 10682.0701224264 \dots > 0.$$

This ambient state is not asserted brake-reachable. It disproves a static first-return barrier and forces any proof of monotonicity to retain the integrated origin of the velocity.

There is a sharp obstruction to closing the first difference using only instantaneous energy and the pair-angular-momentum signs. At a fixed noncollinear shape, let M_v be the translation-reduced Jacobi mass matrix and let the rows of C evaluate L, ℓ_{23}, ℓ_{31} on Jacobi velocity. Prescribing $(L, \ell_{23}, \ell_{31}) = a(0, 1, -x)$ and minimizing kinetic energy gives

$$2K_{\min} = \frac{a^2}{R^2} F, \quad F = (0, 1, -x)(CM_v^{-1}C^T)^{-1}(0, 1, -x)^T > 0,$$

where $R = r_{12}$. This Gram formula is exact. On the pre-syzygy sign cone, write $x = (m_2/m_1)h$, $0 < h < 1$. Zero total angular momentum then gives $-\ell_{12} = a(1-h)/m_1$. If

$$C_1 = \frac{(1-h)^2}{m_1^2} - \frac{1}{p^3} > 0, \quad p = r_{23}/R,$$

the optimal scale-independent energy upper margin for the first gap is

$$\mathcal{M}_1 = \mathfrak{g}_{12} - \mathfrak{g}_{23} + \frac{2U_{\text{sh}}C_1}{F}.$$

This expression is not negative on the whole cone. In the preceding right/obtuse cube, the exact interior point

$$(s, w, v, h) = \left(\frac{99}{100}, 10^{-6}, \frac{99}{100}, 10^{-6}\right)$$

has $C_1 > 0$ and, by exact $\mathbb{Q}(\sqrt{2})$ arithmetic,

$$\mathcal{M}_1 = 93.2280669052696428 \dots > 0.$$

Taking the overall scale sufficiently small realizes the adverse sign at energy $-U_0$ with the minimizing velocity. This state is not asserted to be brake-reachable. Rather, it proves that current shape, energy, zero angular momentum, and the three pair-angular-momentum signs cannot establish the cone; integrated torque history or an equivalent reachability condition is essential.

On the collision-free ordered-syzygy face, however, the first gap does have the desired strict acceleration. Write $(m_1, m_2, m_3) = (m, n, 1)$, normalize $r_{12} = 1$, and suppose body 3 lies between

bodies 1 and 2, so that $q = r_{31}$ and $p = r_{23} = 1 - q$. With $a = \ell_{23} > 0$ and $\eta = (m/n)(-\ell_{31}/\ell_{23})$, transverse momentum and zero total angular momentum give

$$\eta = \frac{q(m+p)}{p(n+q)}, \quad -\frac{\ell_{12}}{a} = \frac{1-\eta}{m}, \quad 2K \geq a^2 \frac{n(mn + mq^2 + np^2)}{mp^2(n+q)^2}.$$

The strict launch pair-angular-momentum signs are equivalent to $0 < q < n/(m+n)$. If the centrifugal coefficient $C_1 = (1-\eta)^2/m^2 - p^{-3}$ is nonpositive, the proved gravitational inequality immediately gives $(r_{12} - r_{23})'' < 0$. Otherwise energy gives the strict upper bound

$$(r_{12} - r_{23})'' < \mathfrak{g}_{12} - \mathfrak{g}_{23} + \frac{2UC_1}{F}, \quad U = mn + \frac{n}{p} + \frac{m}{q}, \quad F = \frac{n(mn + mq^2 + np^2)}{mp^2(n+q)^2}.$$

Put $u = (\sqrt{2}-1)v$ and $q = nz/(m+n)$. After clearing a manifestly positive denominator, the right side has a bidegree-(20, 6) numerator in (v, z) . Its 147 exact $\mathbb{Q}(\sqrt{2})$ Bernstein coefficients consist of 133 negative and 14 zero values. Hence the numerator is strictly negative for $0 < v, z < 1$, and therefore

$$\boxed{(r_{12} - r_{23})'' < 0}$$

at every such ordered syzygy. This closes the first-gap inequality on the terminal face, but neither proves persistence of the ordered cone nor controls the second gap.

The complementary second-gap statement is false under the same instantaneous hypotheses, even with the first-crossing orientation imposed. At $u = 1/3$, take

$$(m, n) = \left(\frac{4}{5}, \frac{3}{5}\right), \quad (r_{12}, r_{23}, r_{31}) = \left(1, \frac{697}{700}, \frac{3}{700}\right).$$

The collinear constraints give $\eta = 419/32759$. Set the radial velocities to zero and choose the transverse amplitude by

$$\ell_{23}^2 = \frac{476958362267}{2867548600}.$$

Then $P = L = 0$, $H = -U_0$, the pair angular momenta have signs $(-, +, -)$, and $\dot{\Delta}_2/\ell_{23} = -292607/131036 < 0$. Nevertheless exact arithmetic gives

$$(r_{23} - r_{31})'' = -\frac{96354167469624287}{1827657180810} < 0.$$

This state is not asserted brake-reachable. It proves that local syzygy geometry, energy, crossing direction, and pair-angular-momentum signs do not control the second gap; an accumulated-history estimate is essential there as well.

The remaining centrifugal terms admit a two-scalar history reduction. Set $R = r_{12}$, $x = r_{23}/R$, $y = r_{31}/R$, $Z = \ell_{23}^2/R$, and retain η . After multiplication by R^2 the two gap contributions are exactly

$$Z \left(\frac{(1-\eta)^2}{m^2} - x^{-3} \right), \quad Z \left(x^{-3} - \frac{n^2\eta^2}{m^2y^3} \right).$$

Thus the sharp interior problem is a reachable barrier for (Z, η) coupled to shape, rather than three independent pair-angular-momentum bounds.

That critical barrier is not pointwise invariant under the natural local constraints. At $u = 1/3$, the exact strict ordered-obtuse state

$$(x, y, \eta) = \left(\frac{499}{500}, \frac{1}{20}, \frac{3}{20}\right)$$

satisfies $k < \eta < k(0)$. Put $Z = (\mathfrak{g}_{23} - \mathfrak{g}_{31})/(-C_2)$, choose the kinetic-minimizing homothetic rate, and choose the positive physical scale from $H = -U_0$. The resulting state has $P = L = 0$, pair-angular-momentum signs $(-, +, -)$, and decreasing area, but exact radical arithmetic gives

$$\frac{d}{ds} \log \frac{Z}{Z_*} = 50.8829734688 \dots > 0.$$

This ambient state is not asserted brake-reachable. It proves that a valid second-gap barrier must encode a stronger correlation from the entire brake history.

At energy $-U_0$, the Jacobi–Maupertuis metric is $2(U - U_0)ds_m^2$. A second-brake segment is a degenerate-boundary geodesic chord. A Morse-index or convexity obstruction remains open.

11 Open proof obligations

The rational and strong real conjectures remain unresolved. The missing theorem must control arbitrarily close collision-free encounters, classify or exclude every double-radial event including Jacobi degeneracies, and be uniform at parameter endpoints. If real intersections remain, an exact arithmetic bridge is still required. In the planar joint-limit family, two transverse collision boundaries are now certified, one for each primary, while the first has a classified local finite-mass collision-or-escape cusp. Global use requires exclusion or certification of the remaining collision zeros and outgoing control on every remaining collision-free component. Differentiated transport through the finite-mass saddle dwell is complete; on the exact tied curve the remaining local condition is the scalar sign/zero problem for $\Delta_{\text{intr}}(B)$. Phase sampling forces infinitely many real classical collisions but does not classify each collision-free component or yield a rational root.

References

- [1] C. Burrau, Numerische Berechnung eines Spezialfalles des Dreikörperproblems, *Astronomische Nachrichten* 195 (1913), 113–118. [doi:10.1002/asna.19131950602](https://doi.org/10.1002/asna.19131950602). [Original scan](#).
- [2] V. Szebehely, Burrau’s problem of three bodies, *Proceedings of the National Academy of Sciences USA* 58 (1967), 60–65. [doi:10.1073/pnas.58.1.60](https://doi.org/10.1073/pnas.58.1.60). [Original article](#), [PostScript facsimile](#).
- [3] E. Strömberg, A new class of periodic solutions in the general problem of three bodies, viz. in the problem of three bodies with all masses and all distances of the same order of magnitude, *Monthly Notices of the Royal Astronomical Society* 80 (1919), 12–22. [Original scan](#).
- [4] J. Zunkley, Ein numerisch gerechneter Spezialfall des allgemeinen Dreikörperproblems in vereinfachter Behandlung, *Astronomische Nachrichten* 272 (1941), 66–76. [Original scan](#).
- [5] V. Szebehely and C. F. Peters, Complete solution of a general problem of three bodies, *Astronomical Journal* 72 (1967), 876–883. [Original article](#).
- [6] V. Szebehely and C. F. Peters, A new periodic solution of the problem of three bodies, *Astronomical Journal* 72 (1967), 1187–1190. [Original article](#).
- [7] E. M. Standish, Jr., New periodic orbits in the general problem of three bodies, in G. E. O. Giacaglia (ed.), *Periodic Orbits, Stability and Resonances*, D. Reidel, Dordrecht (1970), 375–381. [doi:10.1007/978-94-010-3323-7_34](https://doi.org/10.1007/978-94-010-3323-7_34).

- [8] K. Tanikawa, H. Umehara, and H. Abe, A search for collision orbits in the free-fall three-body problem I. Numerical procedure, *Celestial Mechanics and Dynamical Astronomy* 62 (1995), 335–362. [doi:10.1007/BF00692285](https://doi.org/10.1007/BF00692285).
- [9] K. Tanikawa, A search for collision orbits in the free-fall three-body problem II, *Celestial Mechanics and Dynamical Astronomy* 76 (2000), 157–185. [doi:10.1023/A:1008313912013](https://doi.org/10.1023/A:1008313912013).
- [10] R. Moeckel, R. Montgomery, and A. Venturelli, From brake to syzygy, *Archive for Rational Mechanics and Analysis* 204 (2012), 1009–1060. [doi:10.1007/s00205-012-0502-y](https://doi.org/10.1007/s00205-012-0502-y). [Author manuscript](#).
- [11] R. Montgomery, Dropping bodies, *The Mathematical Intelligencer* (2023). [doi:10.1007/s00283-022-10252-4](https://doi.org/10.1007/s00283-022-10252-4). [Author preprint \(2022\)](#).
- [12] X. Li and S. Liao, Collisionless periodic orbits in the free-fall three-body problem, *New Astronomy* 70 (2019), 22–26. [doi:10.1016/j.newast.2019.01.003](https://doi.org/10.1016/j.newast.2019.01.003).
- [13] I. Hristov, R. Hristova, V. Dmitrašinović, and K. Tanikawa, Three-body periodic collisionless equal-mass free-fall orbits revisited, *Celestial Mechanics and Dynamical Astronomy* 136 (2024), article 7. [doi:10.1007/s10569-023-10177-w](https://doi.org/10.1007/s10569-023-10177-w). [Author preprint](#).
- [14] A. Rantala, P. Pihajoki, M. Mannerkoski, P. H. Johansson, and T. Naab, MSTAR—a fast parallelized algorithmically regularized integrator with minimum spanning tree coordinates, *Monthly Notices of the Royal Astronomical Society* 492 (2020), 4131–4149. [Publisher article](#), Section 5.2.
- [15] A. S. Chitan, A. Mylläri, and S. Haque, Relativistic effects on triple black holes: Burrau’s problem revisited, *Monthly Notices of the Royal Astronomical Society* 509 (2022), 1919–1928. [Publisher article](#). [Author preprint](#).
- [16] A. Chitan, A. Mylläri, and M. Valtonen, The influence of spin in black hole triplets, *Astronomy and Computing* 51 (2025), 100933. [doi:10.1016/j.ascom.2025.100933](https://doi.org/10.1016/j.ascom.2025.100933).
- [17] E. M. Standish, Sufficient conditions for escape in the three-body problem, *Celestial Mechanics* 4 (1971), 44–48. [doi:10.1007/BF01230319](https://doi.org/10.1007/BF01230319).
- [18] J. Yoshida, Improved criteria for hyperbolic–elliptic motion in the general three-body problem, *Publications of the Astronomical Society of Japan* 24 (1972), 391–408.
- [19] L. F. Bakker and S. C. Simmons, A separating surface for Sitnikov-like $n + 1$ -body problems, *Journal of Differential Equations* 258 (2015), 3063–3087.
- [20] R. McGehee, A stable manifold theorem for degenerate fixed points with applications to celestial mechanics, *Journal of Differential Equations* 14 (1973), 70–88.